University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

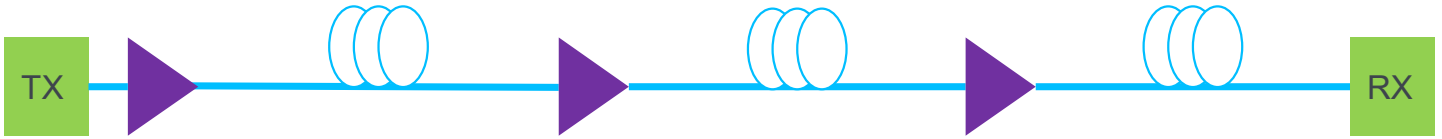
Lecture 12

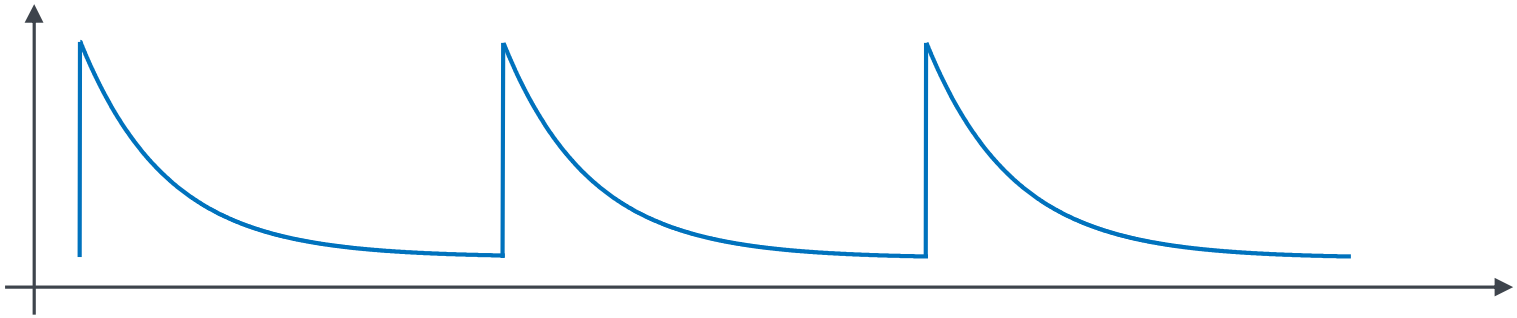


Prof. Dr.-Ing. Georg Rademacher

Power profile in amplified transmission link

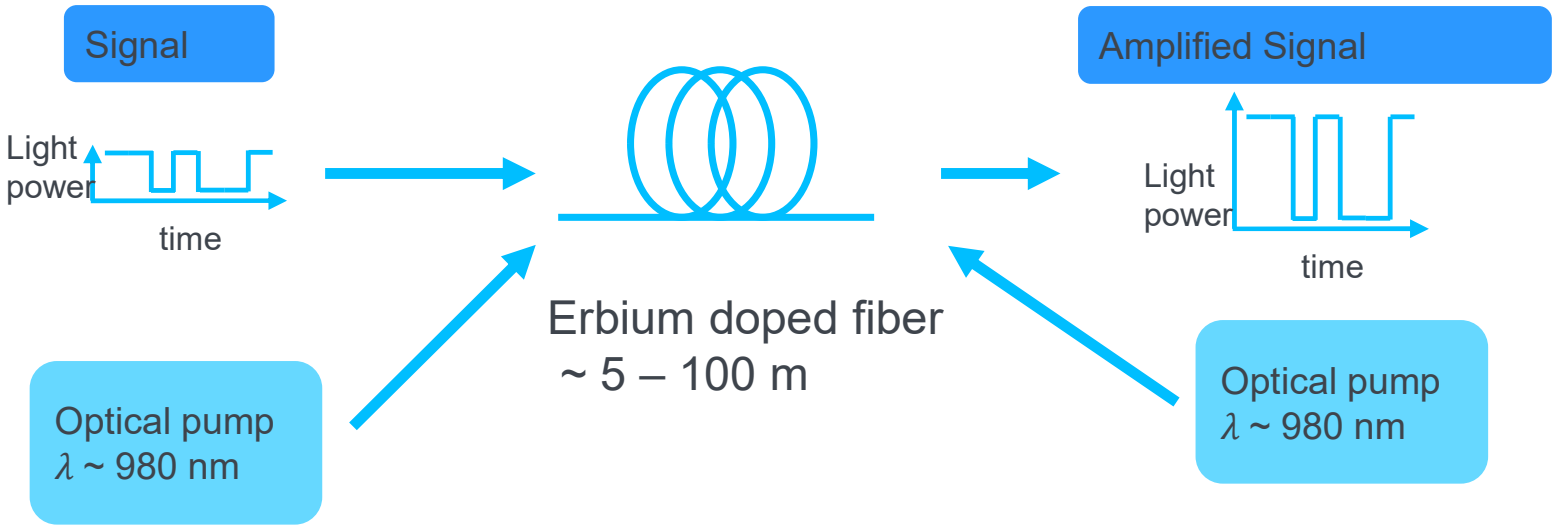
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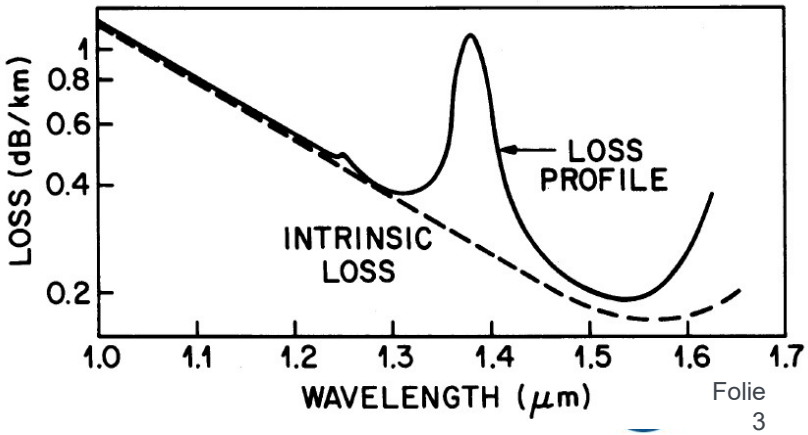




Erbium doped fiber amplifier (EDFA)

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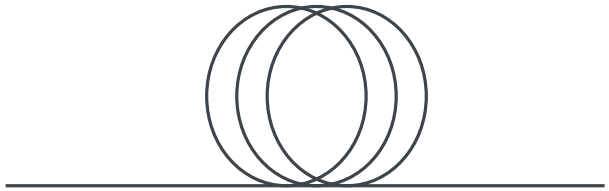
Noise in long distance transmission

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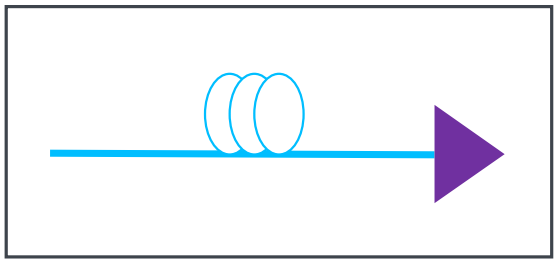
Noise in long distance transmission

University of Stuttgart
Germany



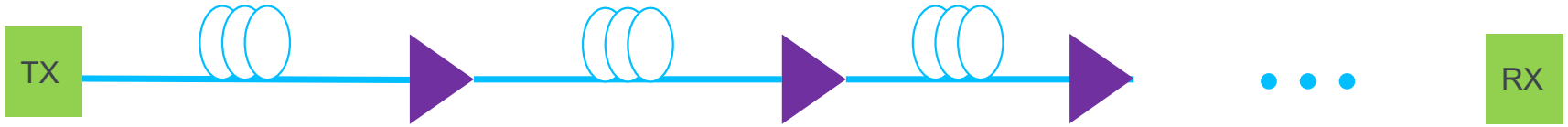
Noise in long distance transmission

University of Stuttgart
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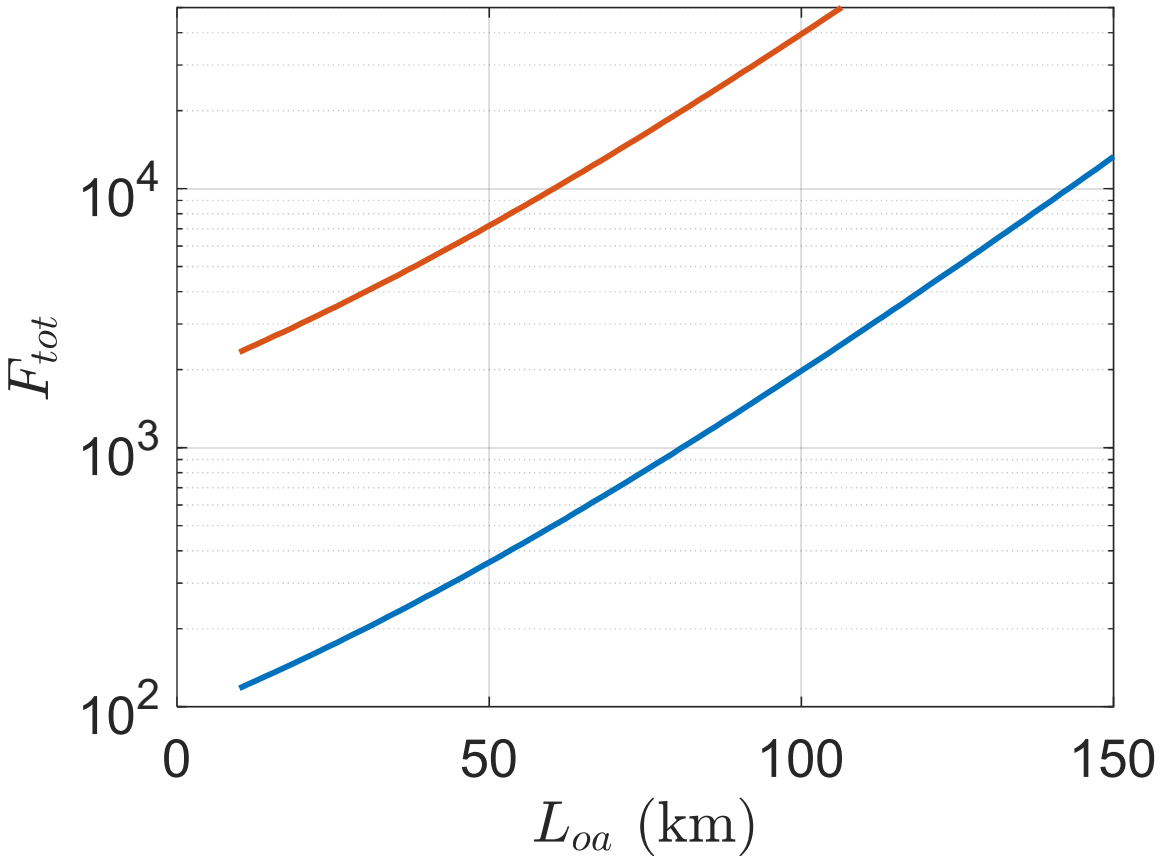
Noise in long distance transmission

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Germany



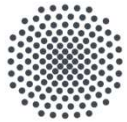
Noise in long distance transmission

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Germany



Optical Signal to Noise Ratio (OSNR)

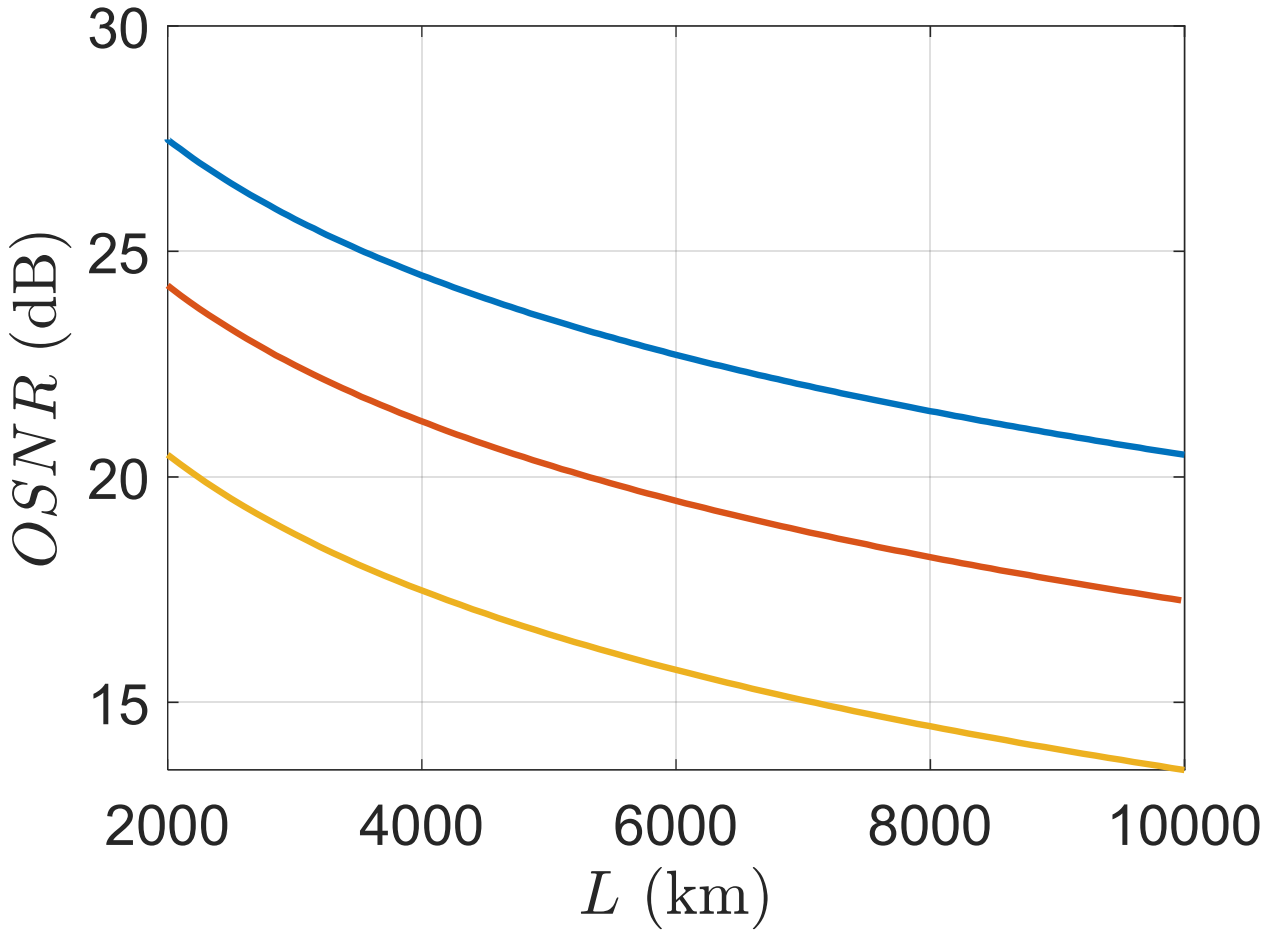
OSNR in long distance transmission



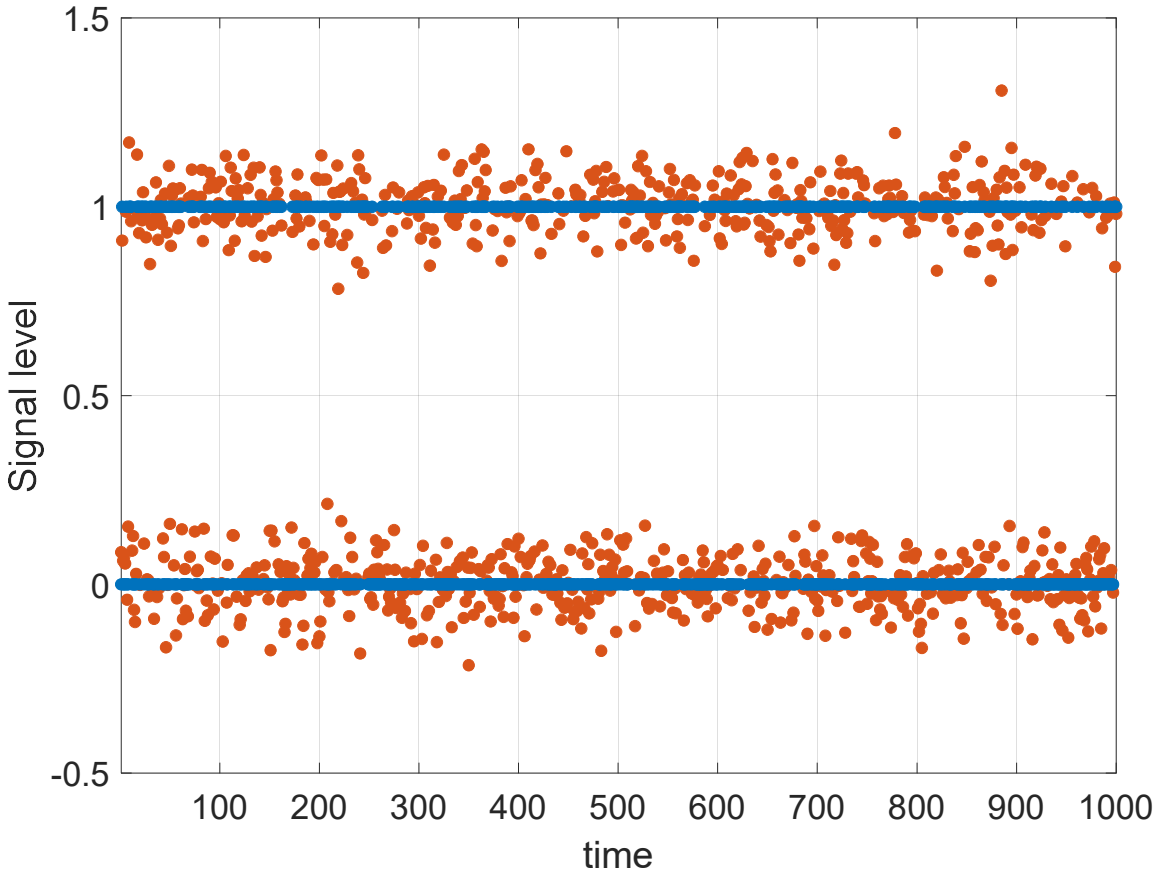
University of Stuttgart
Germany

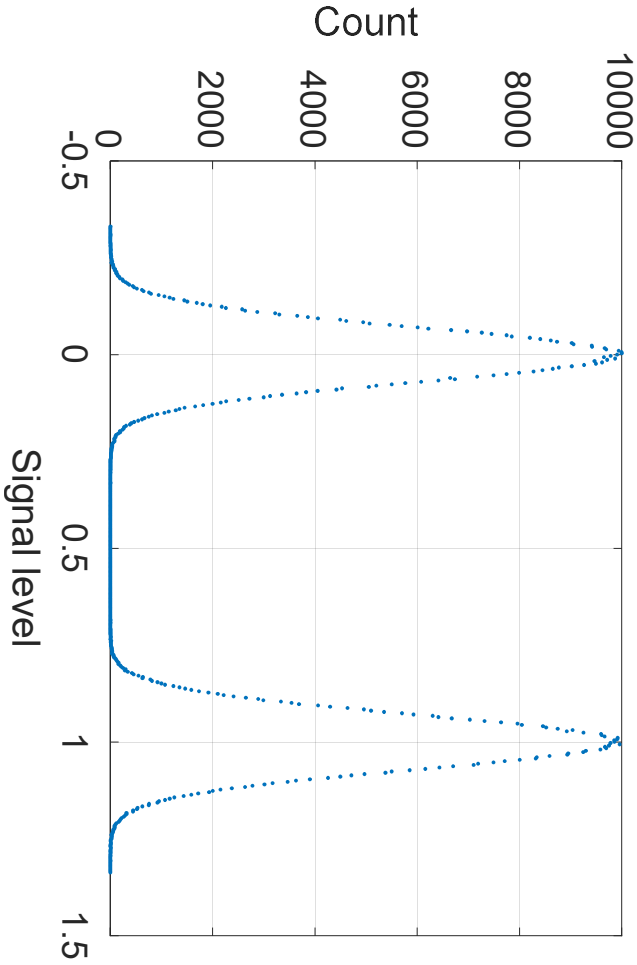
OSNR in long distance transmission

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Germany

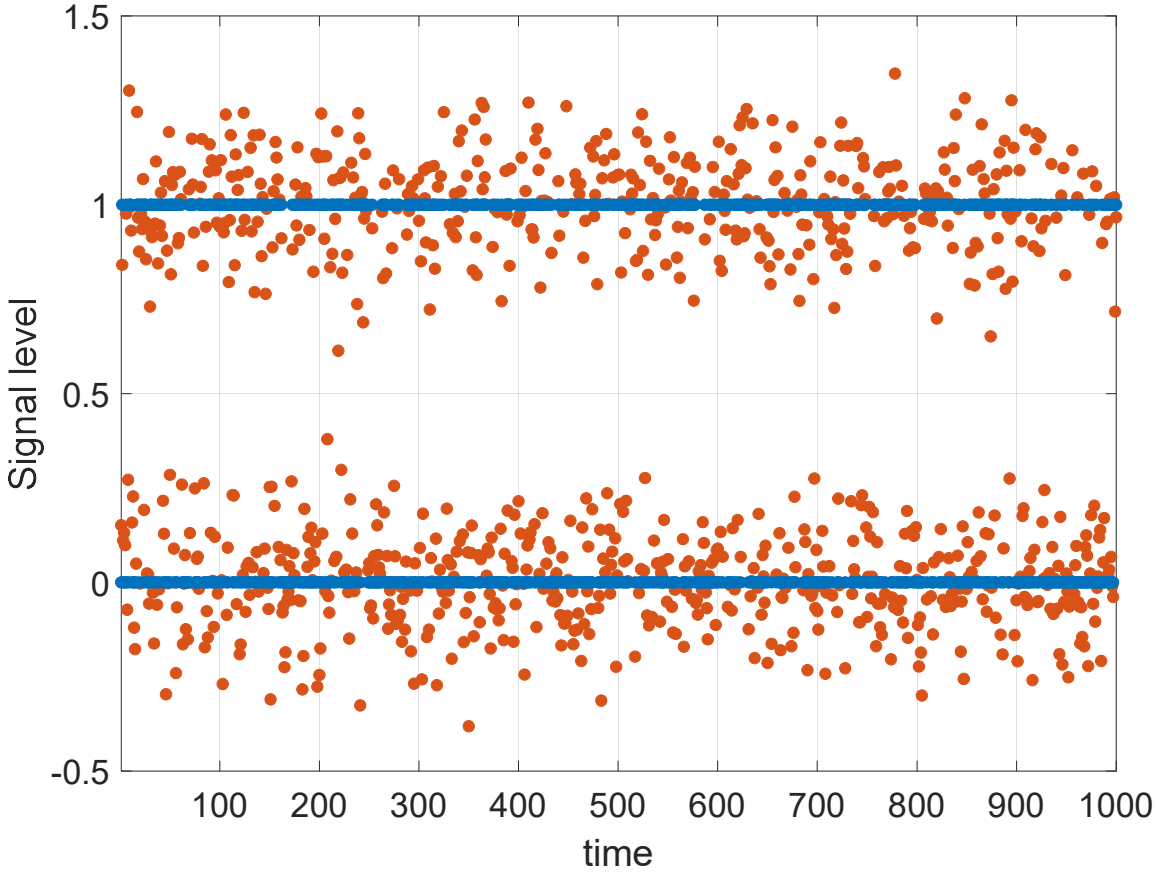


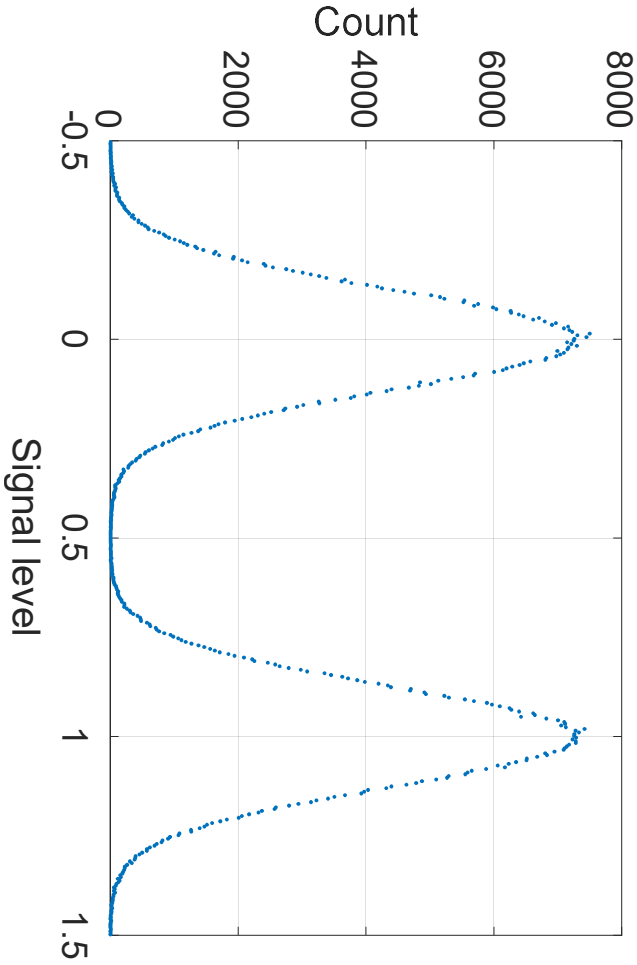
Bit-Error Rate (BER)



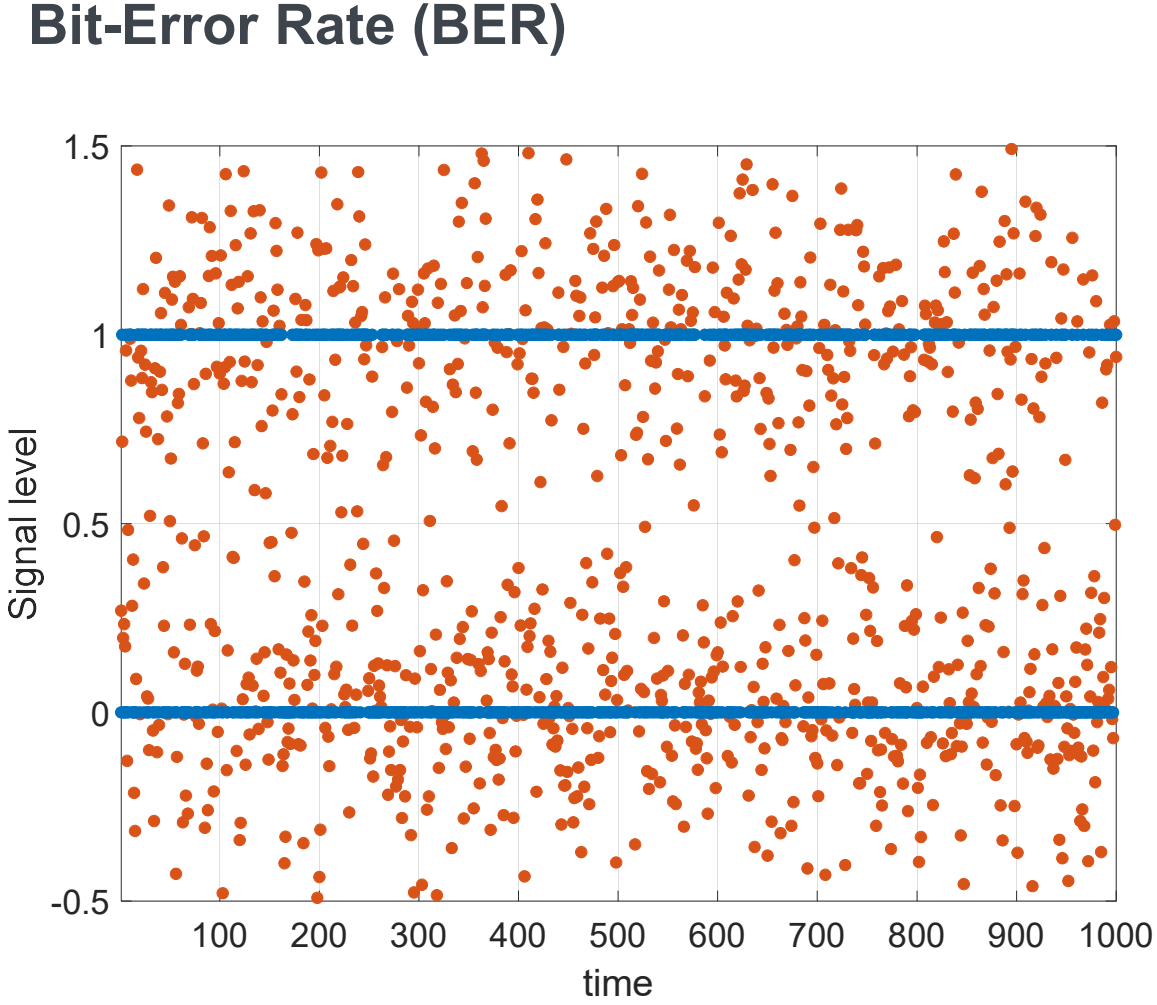


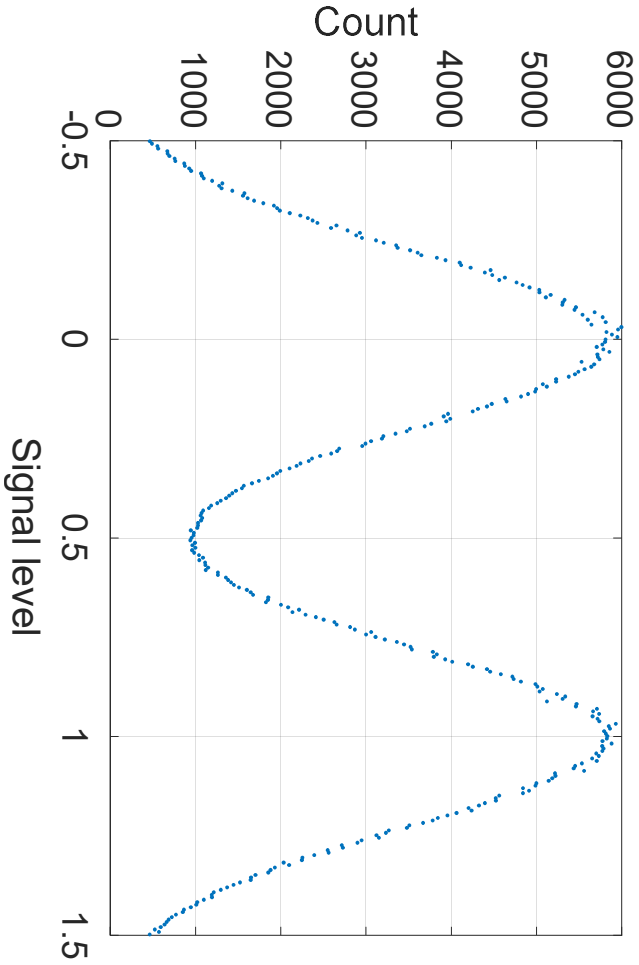
Bit-Error Rate (BER)





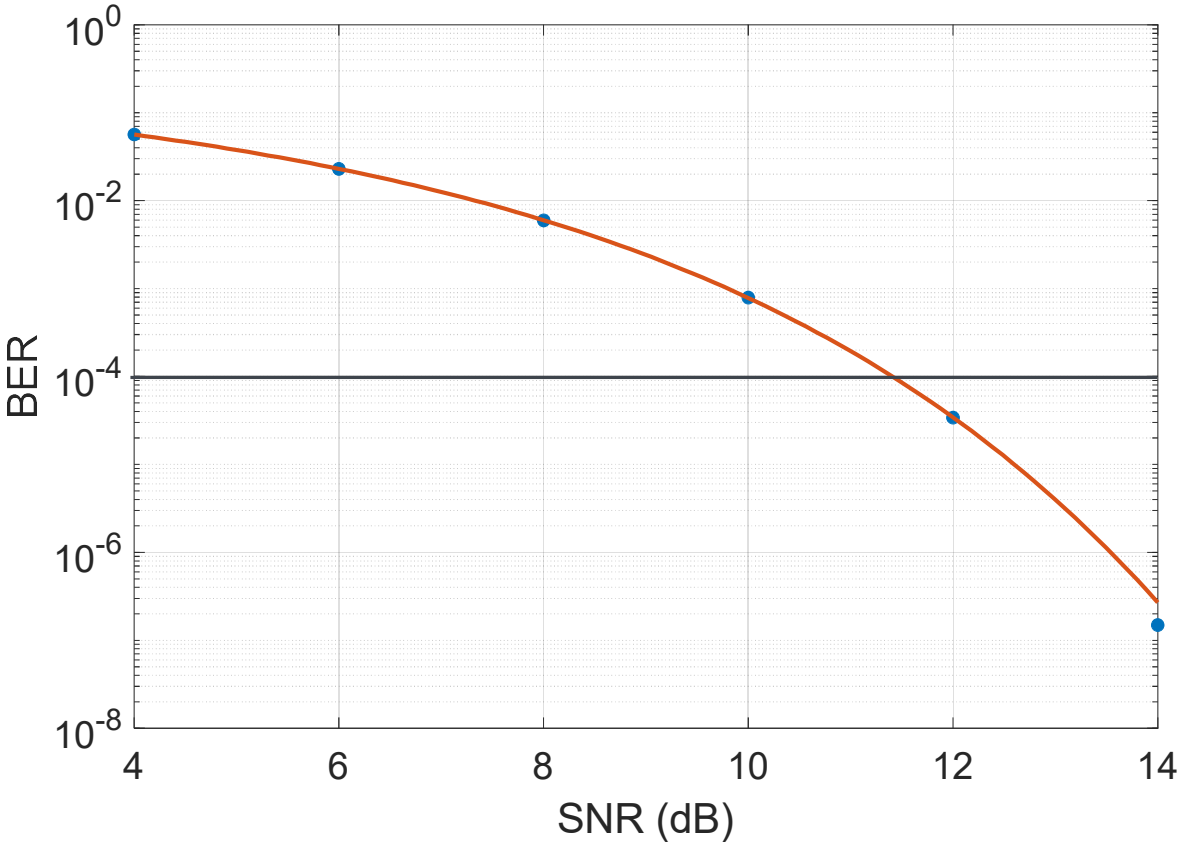
Bit-Error Rate (BER)





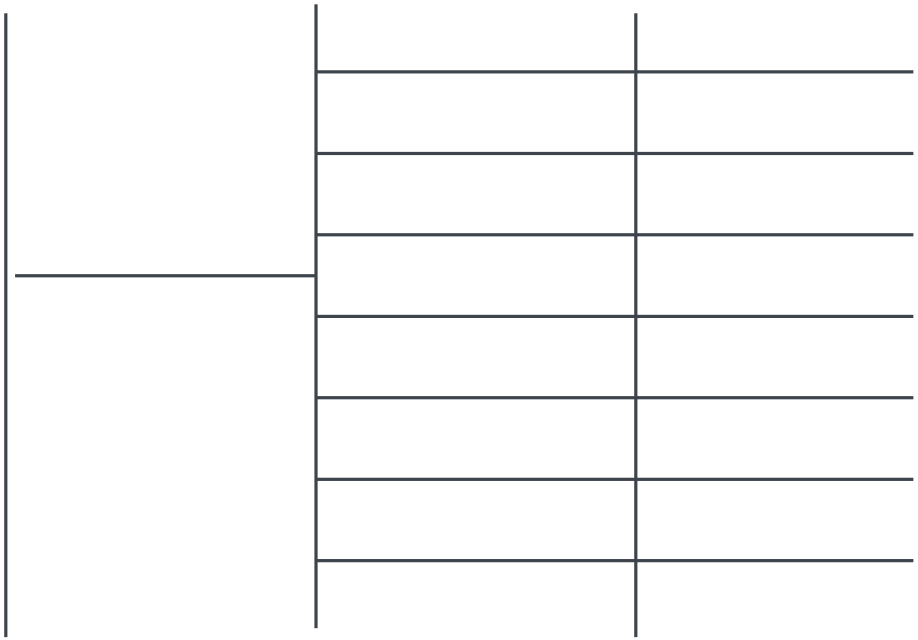
BER vs SNR for OOK

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Forward Error Correction (FEC)

Payload bit Msg



Forward Error Correction (FEC)

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Information-Theoretic Tools for Optical Communications Engineers

(Invited Tutorial)

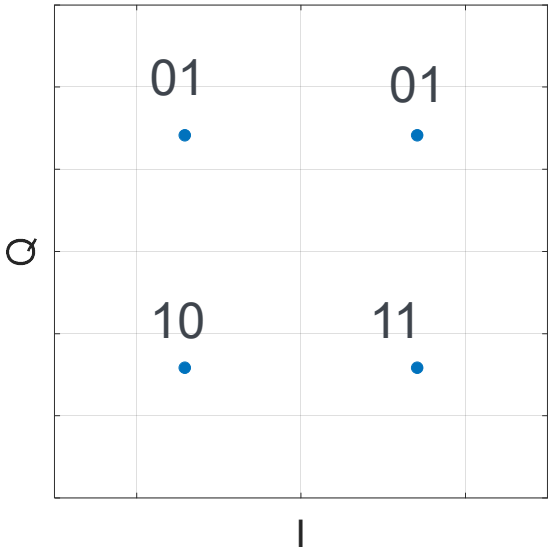
Erik Agrell
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Marco Secondini
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Scuola Superiore Sant'Anna
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marco.secondini@sssup.it

TABLE I
HARD-DECISION FEC THRESHOLDS OF CERTAIN CODES RECOMMENDED FOR OPTICAL COMMUNICATIONS

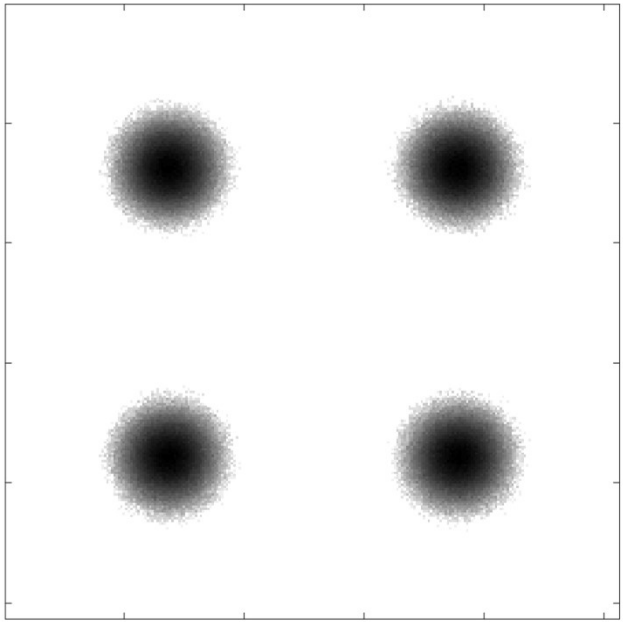
OH	Code type	Source	post-FEC BER	p
2.7%	Reed–Solomon (RS) (528,514) "KR4"	[26][a]	10^{-15}	$2.18 \cdot 10^{-5}$
5.8%	RS(544,514) "KP4"	[26][a]	10^{-15}	$2.26 \cdot 10^{-4}$
6.7%	RS(255,239)	[27][a]	10^{-15}	$8.27 \cdot 10^{-5}$
6.7%	Bose–Chaudhuri–Hocquenghem (BCH)+BCH	[28, App. I.3]	10^{-15}	$3.15 \cdot 10^{-3}$

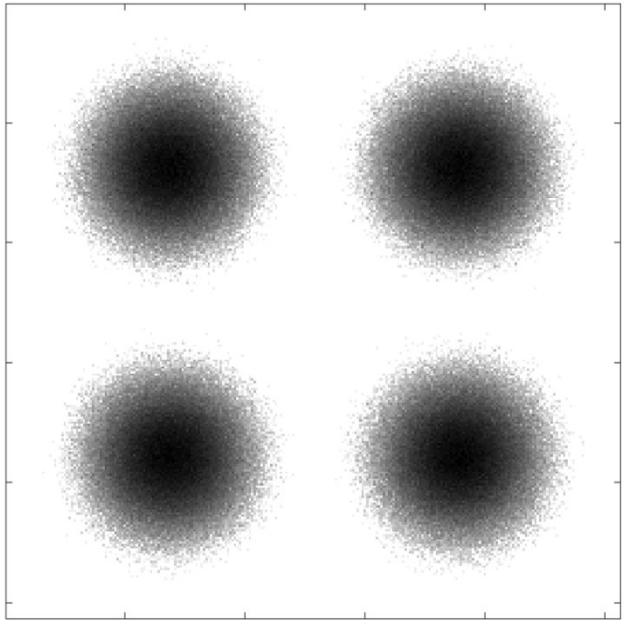
Coherent Signals

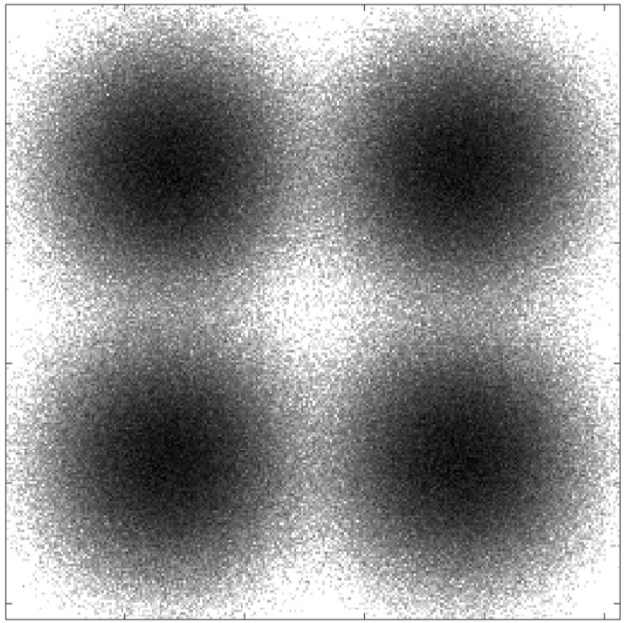


Coherent Signals

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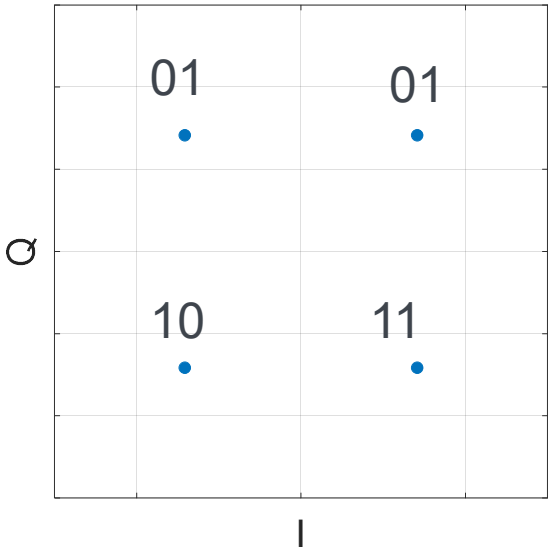


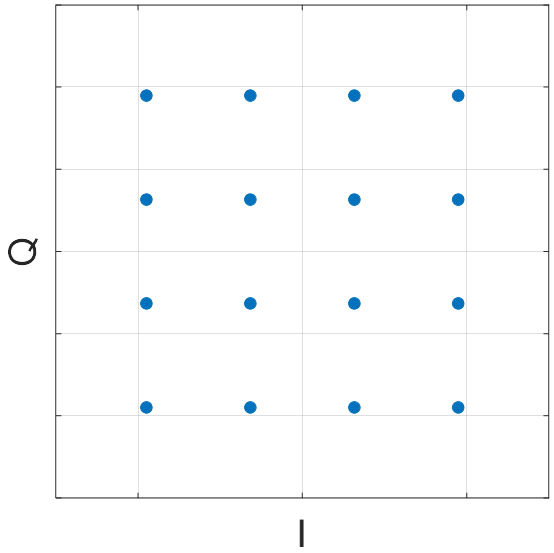


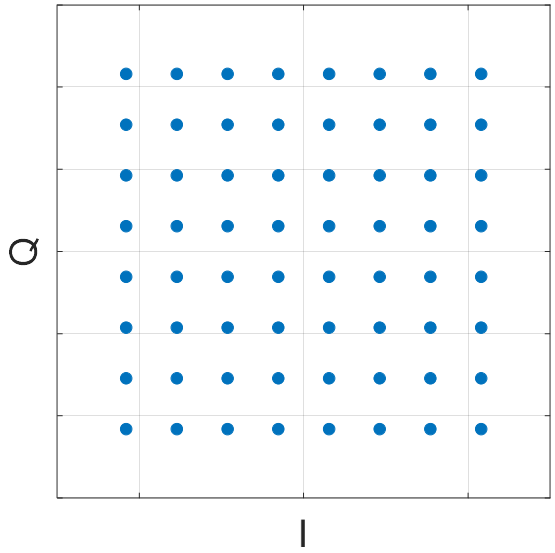


Coherent Signals

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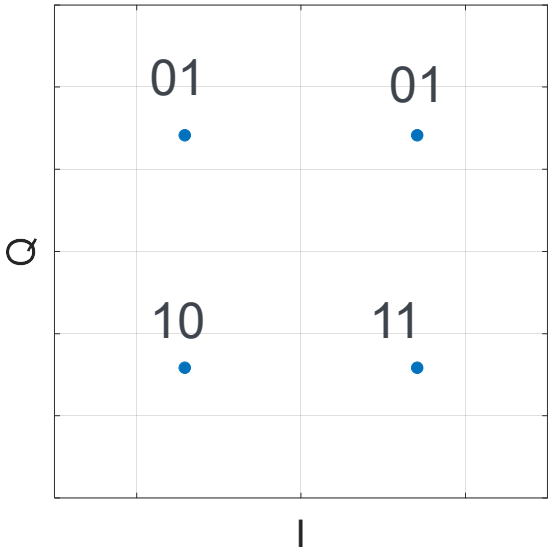


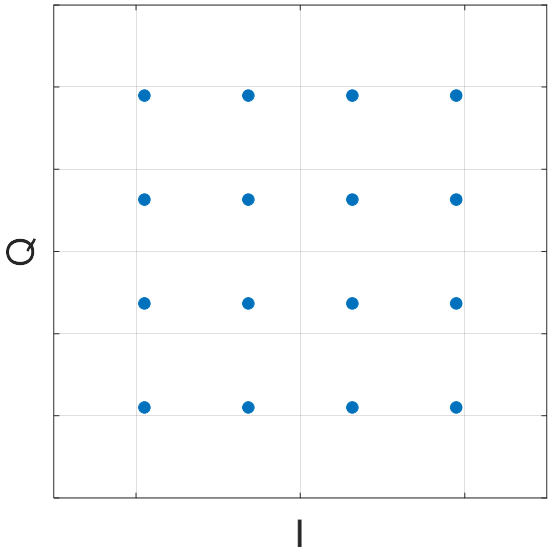


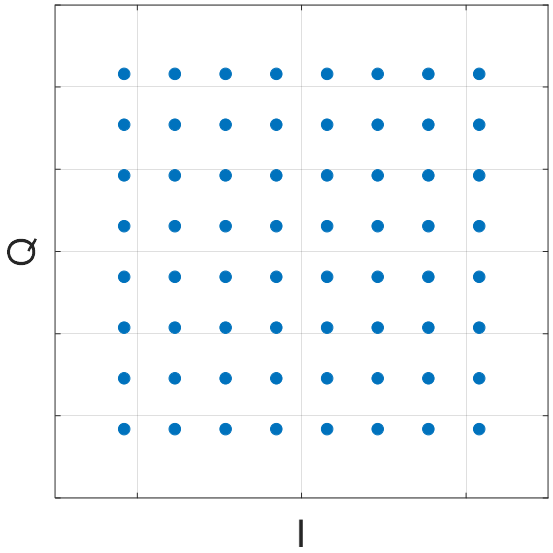


Coherent Signals

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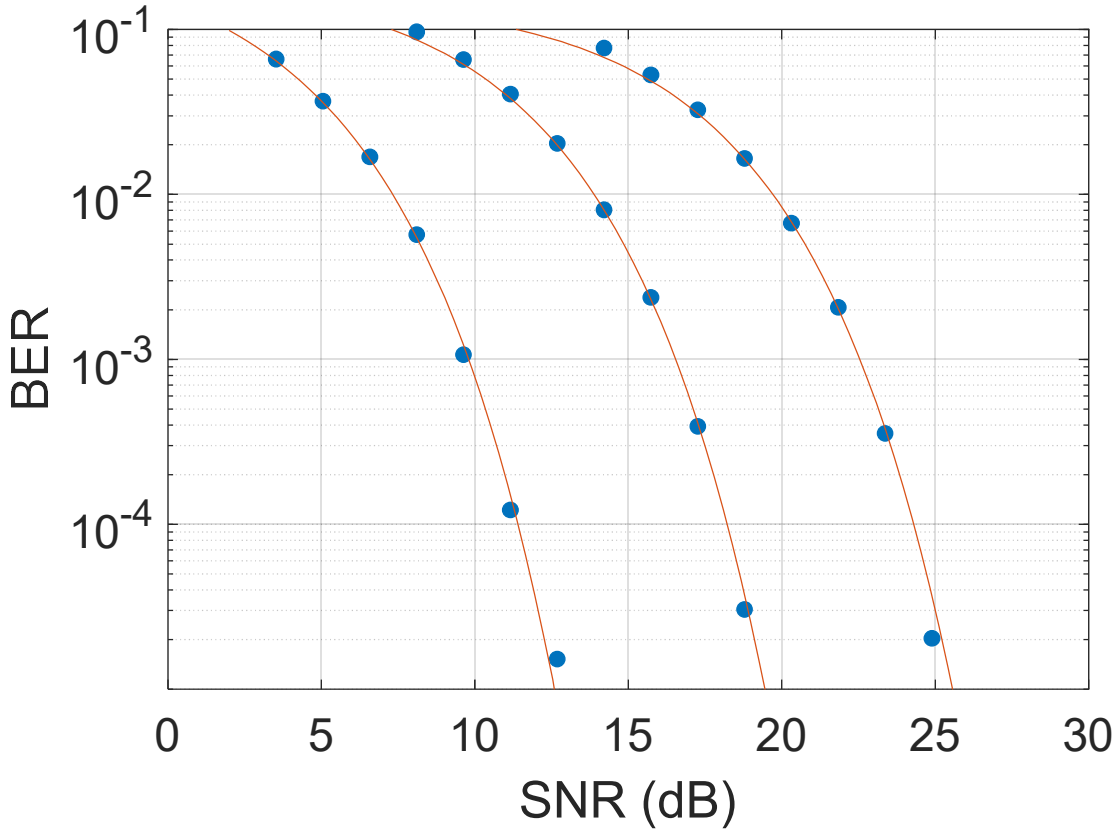






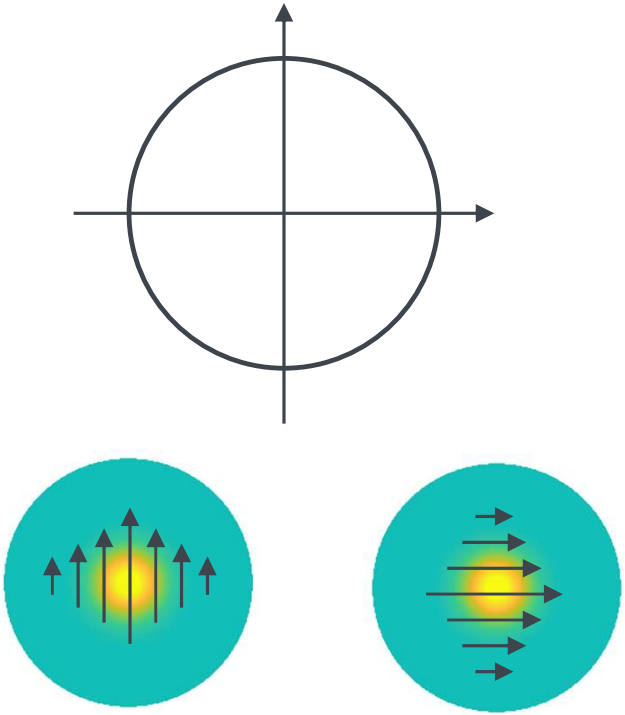
BER vs. SNR

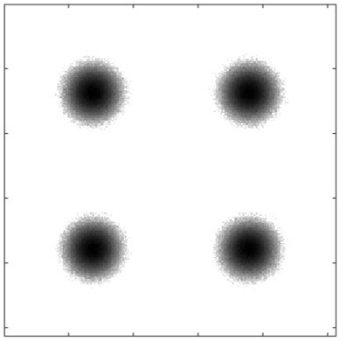
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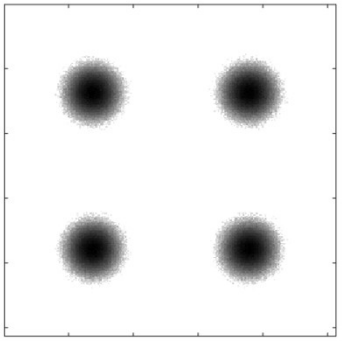


Polarization multiplex (Pol Mux)

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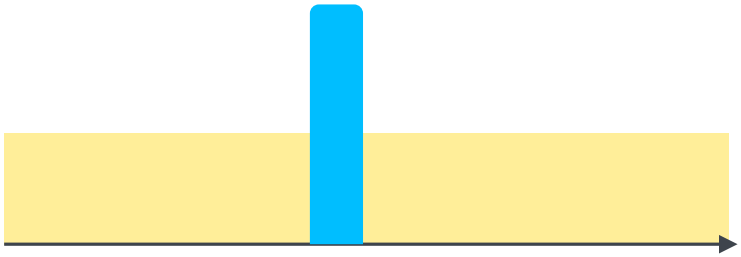






SNR vs. OSNR

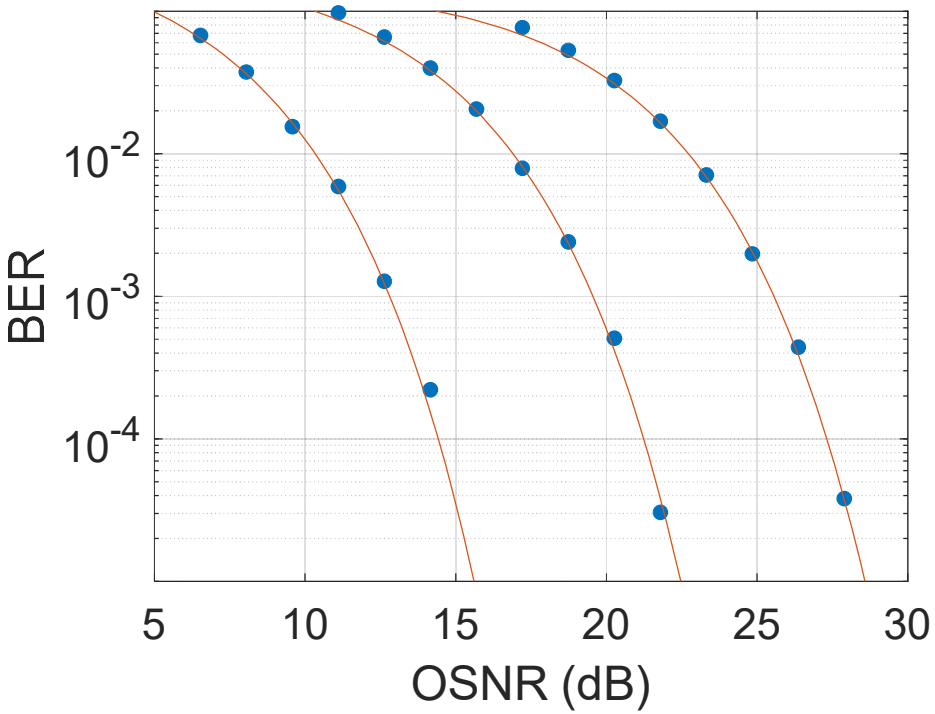
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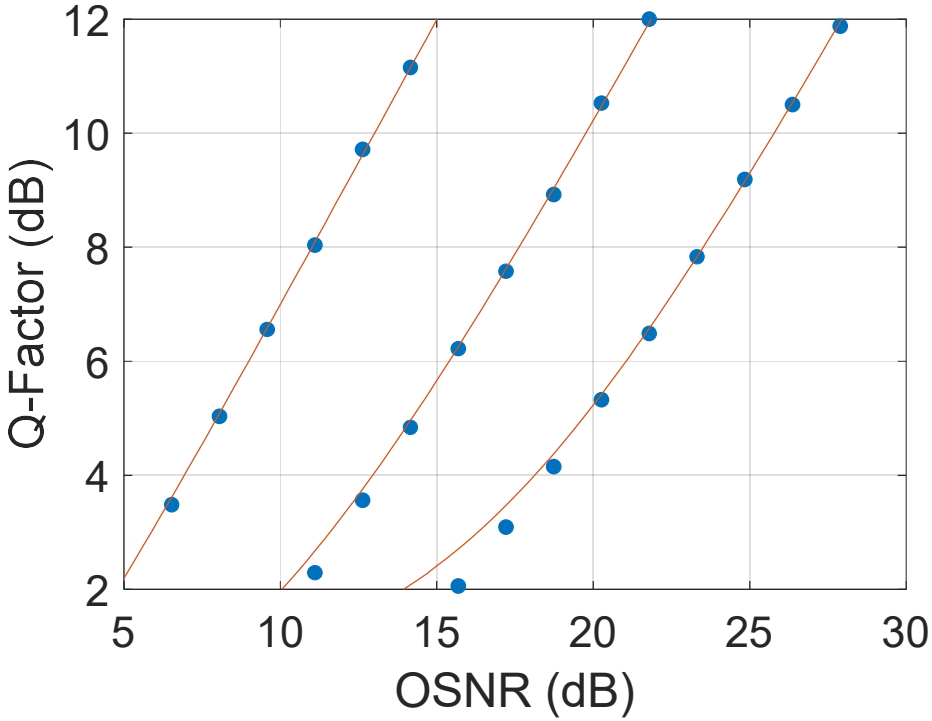




Coherent Signals

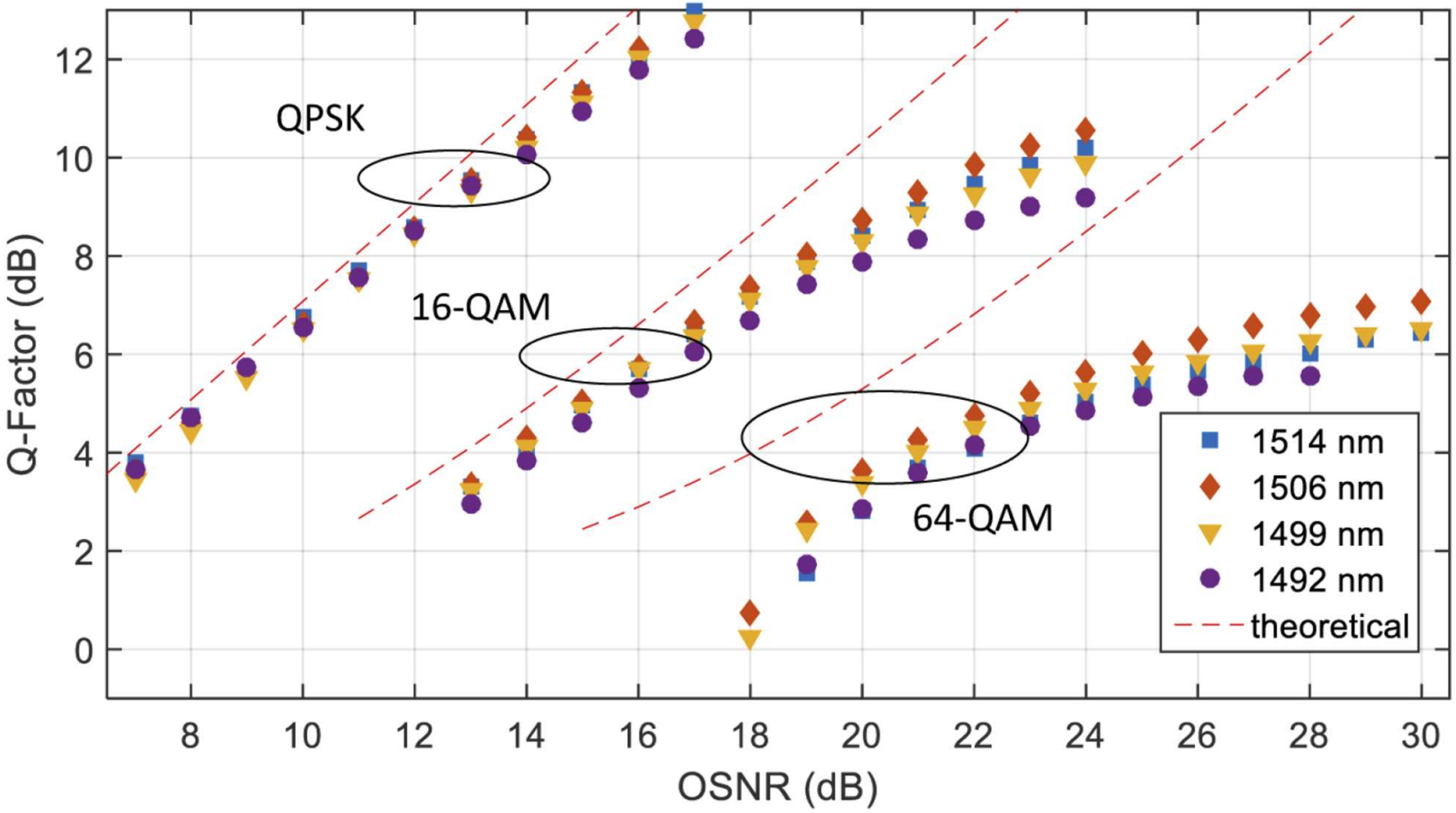
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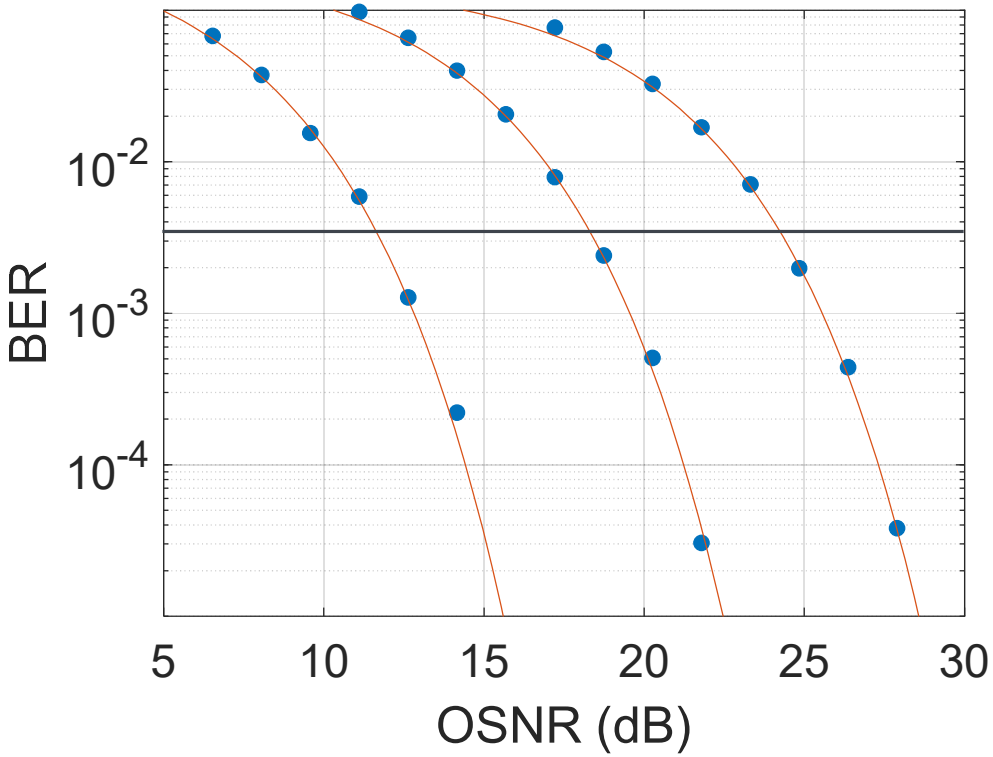
Measured performance

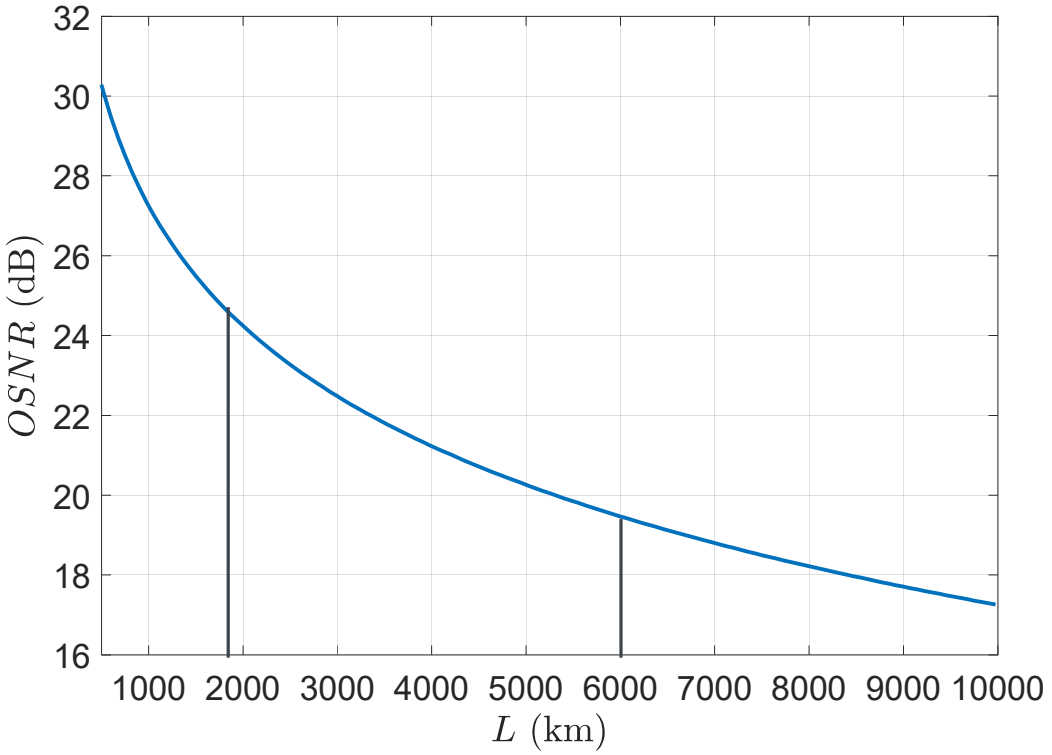
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Long distance transmission

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Pulse propagation in SMF

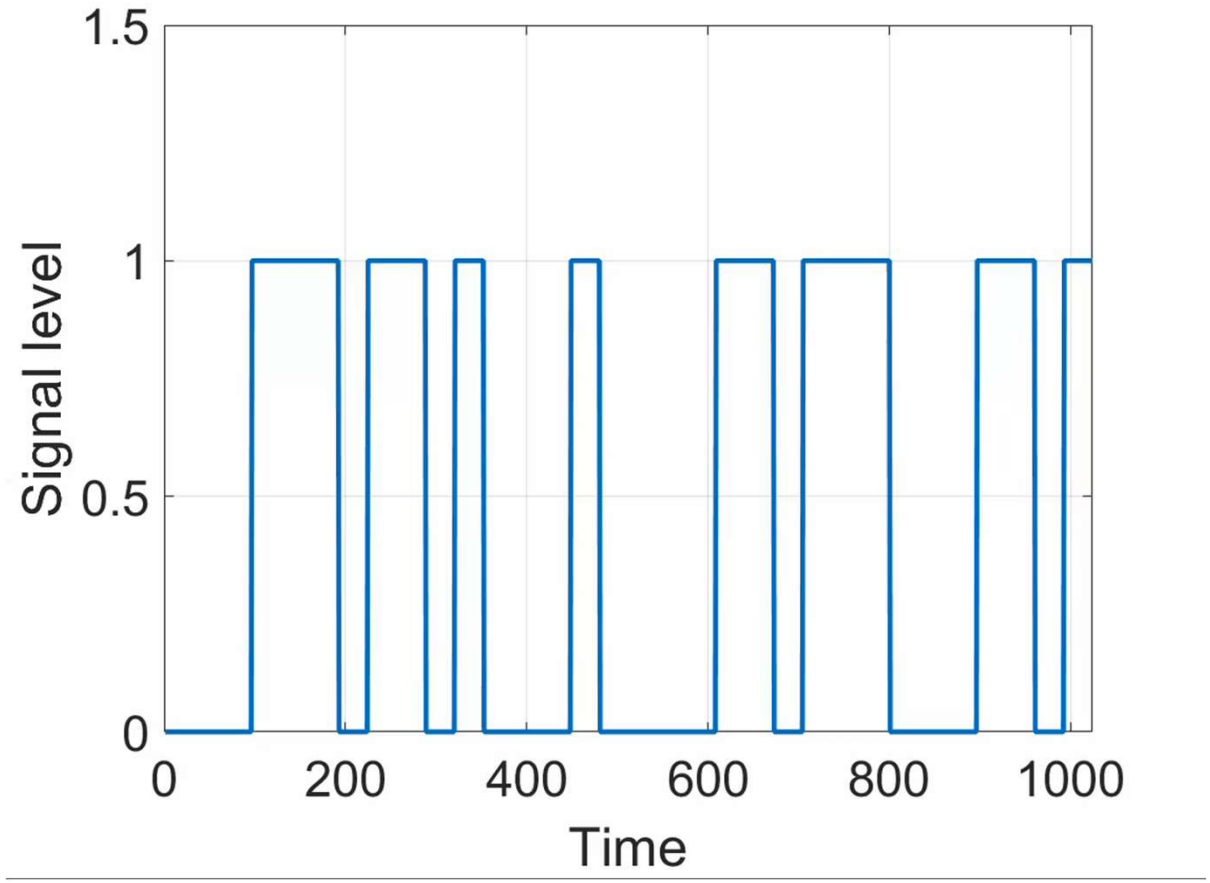
Pulse propagation in SMF

Pulse propagation in SMF

Pulse propagation in SMF

Pulse propagation in SMF

Pulse propagation in SMF


1.5
1
0.5
0
Signal level
0
200
400
600
800
1000
Time

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Thank you!

Prof. Dr.-Ing. Georg Rademacher

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phone +49 (0) 711 6856 7901
www. int.uni-stuttgart.de/

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Advanced optical fiber communications

Wintersemester 2025/26

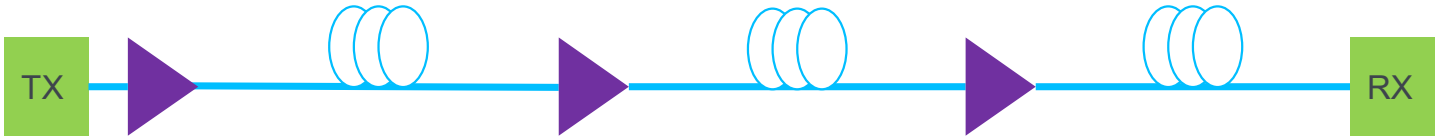
Lecture 12

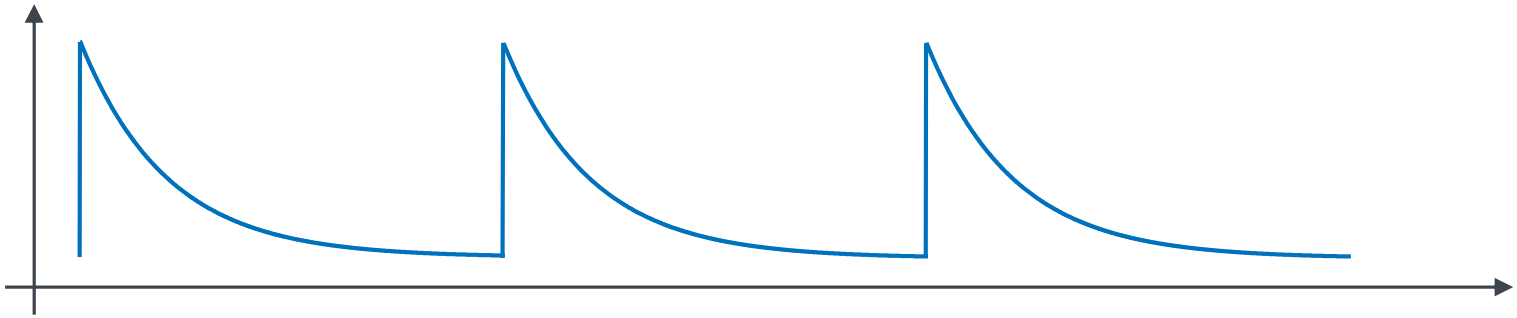


Prof. Dr.-Ing. Georg Rademacher

Power profile in amplified transmission link

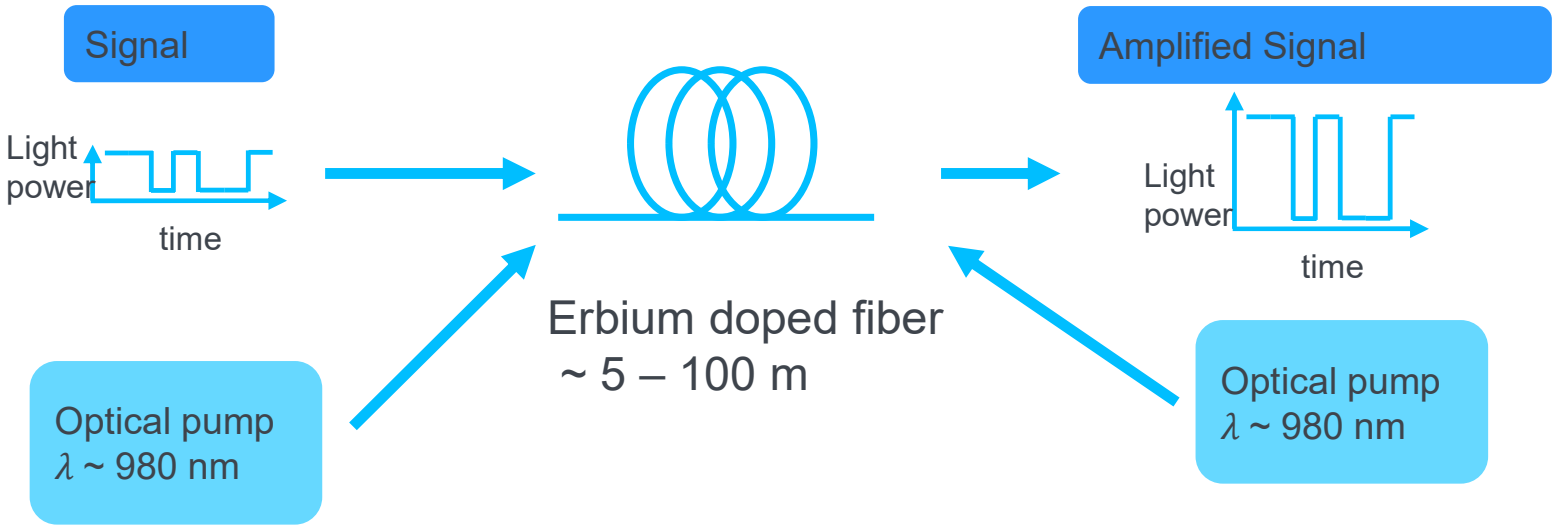
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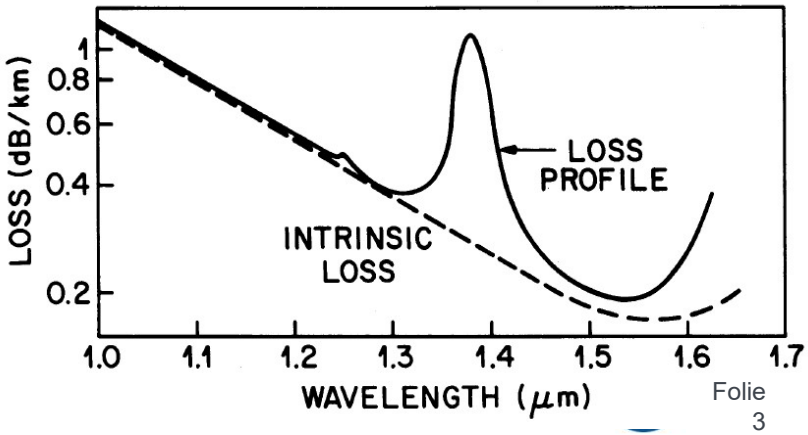




Erbium doped fiber amplifier (EDFA)

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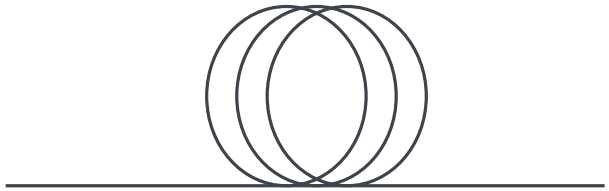
Noise in long distance transmission

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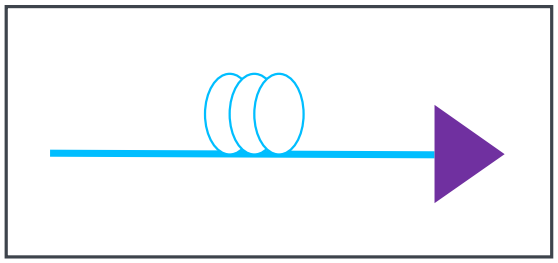
Noise in long distance transmission

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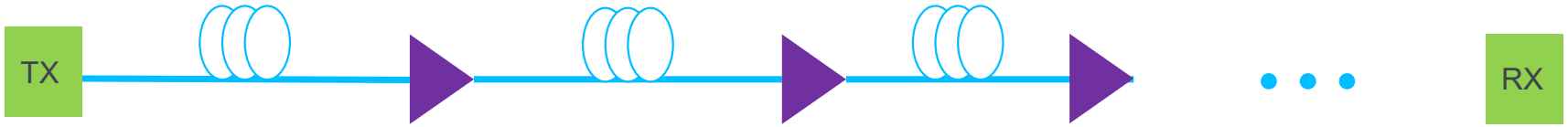
Noise in long distance transmission

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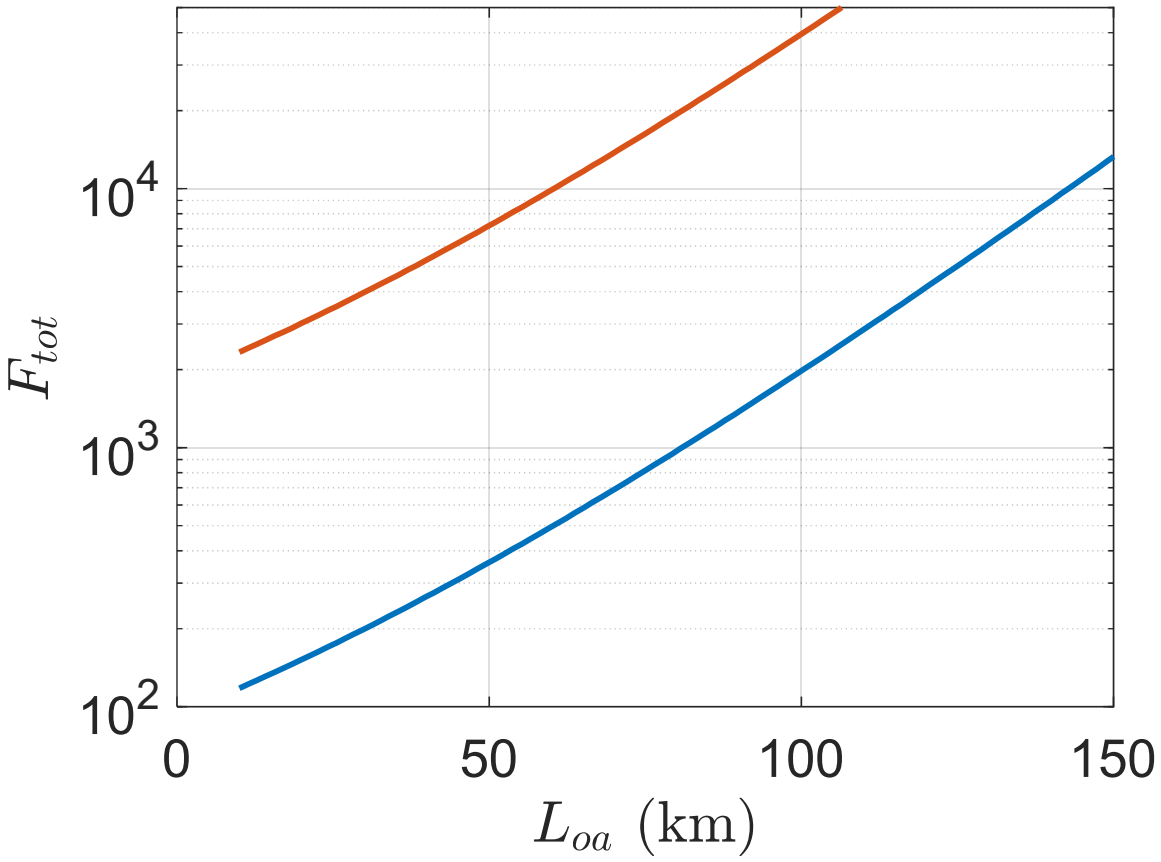
Noise in long distance transmission

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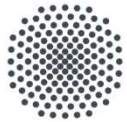
Noise in long distance transmission

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Optical Signal to Noise Ratio (OSNR)

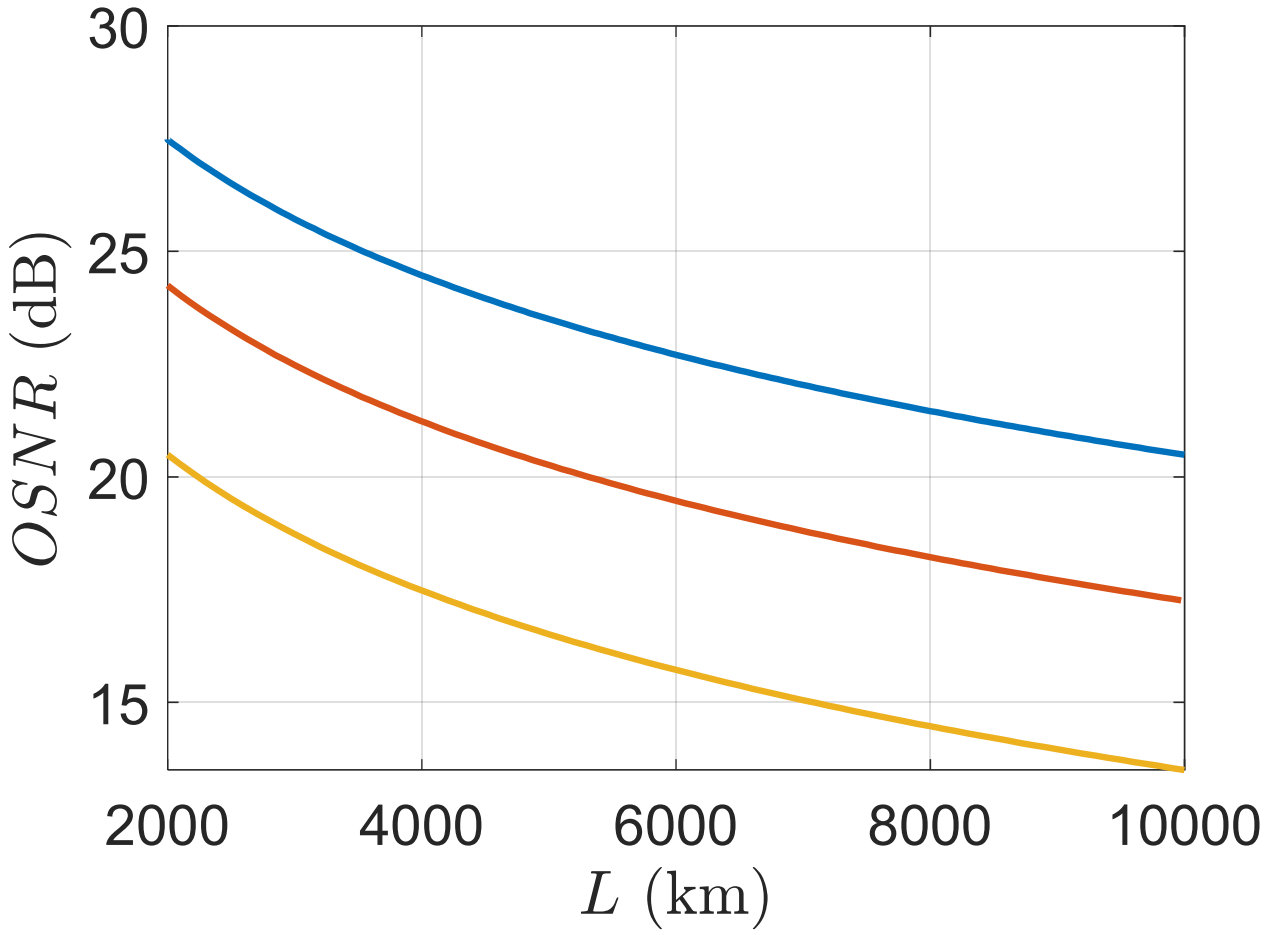
OSNR in long distance transmission



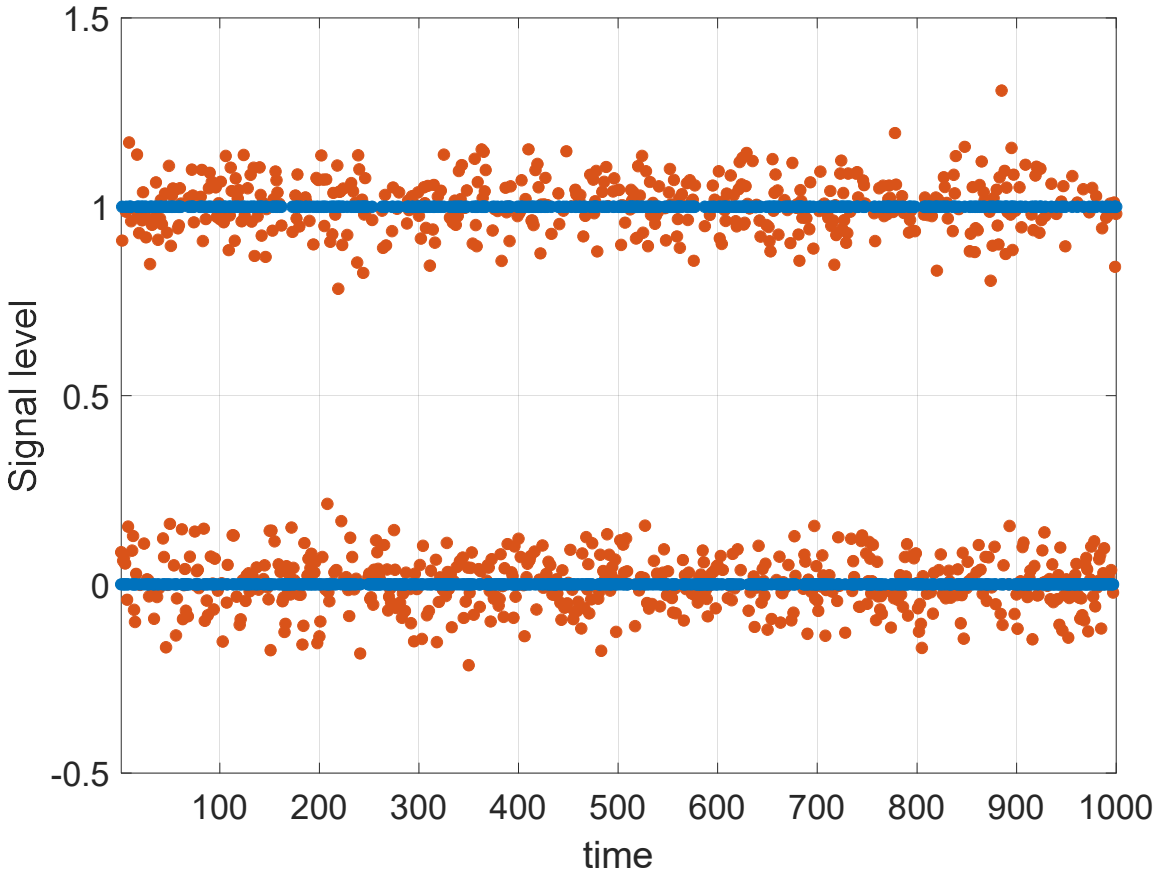
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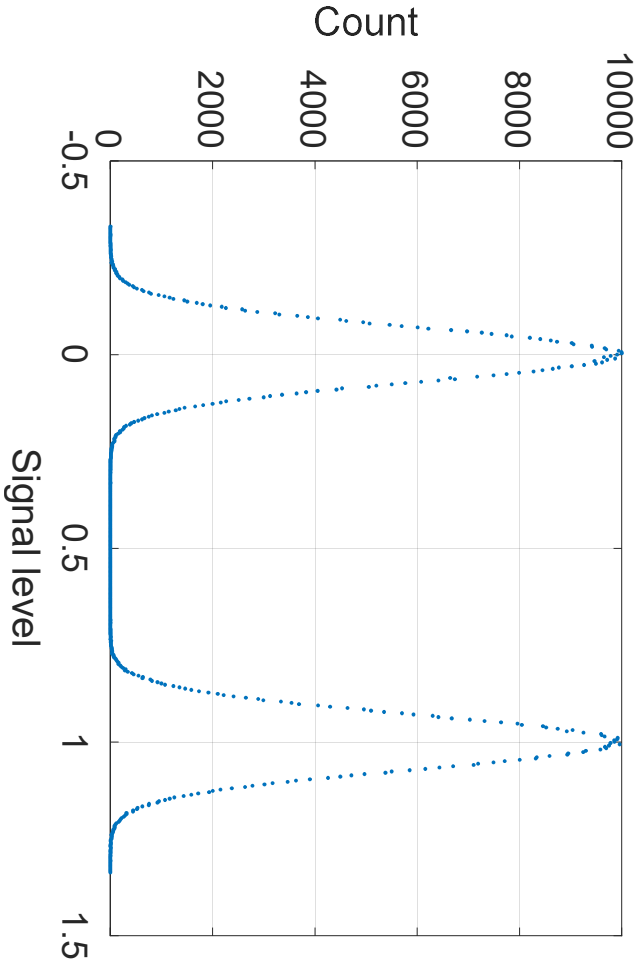
OSNR in long distance transmission

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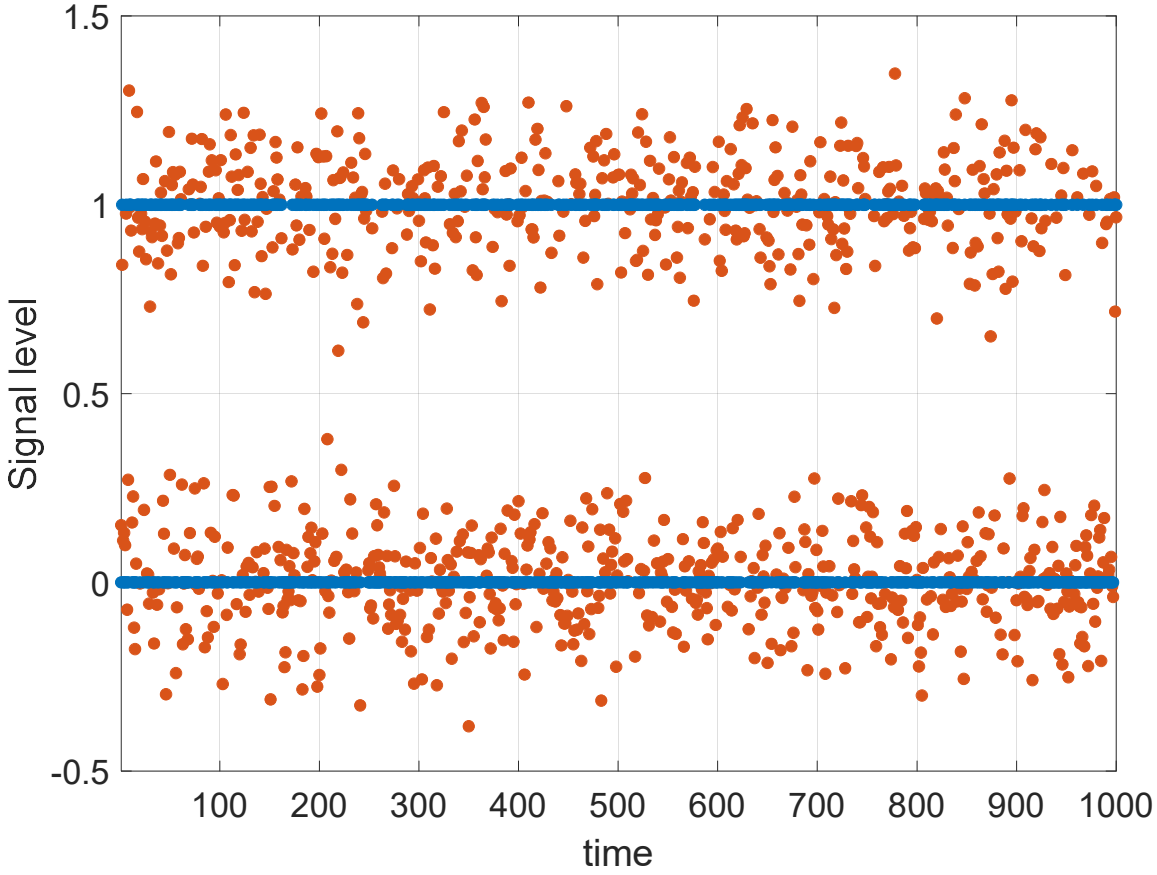


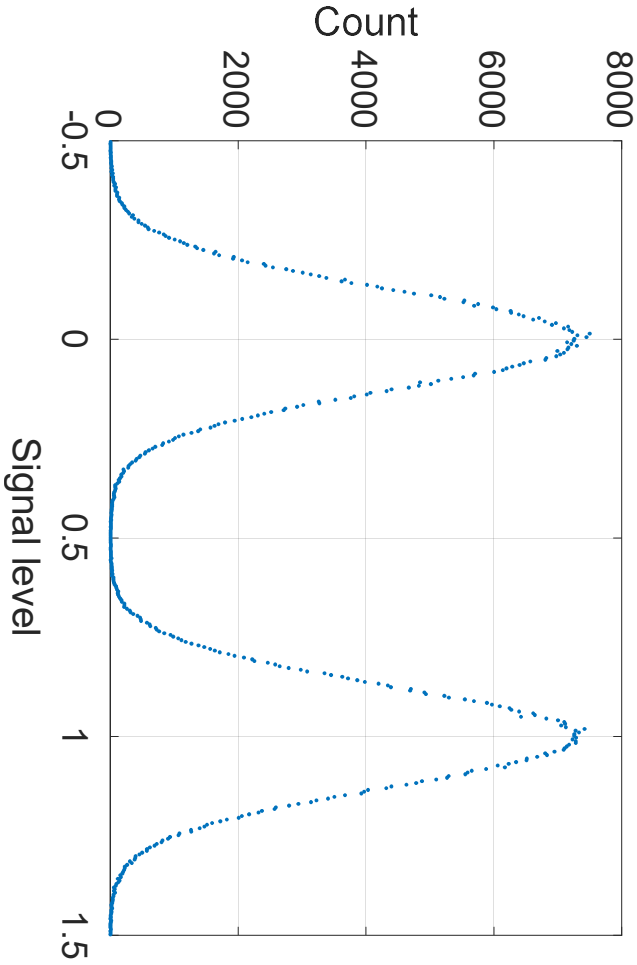
Bit-Error Rate (BER)



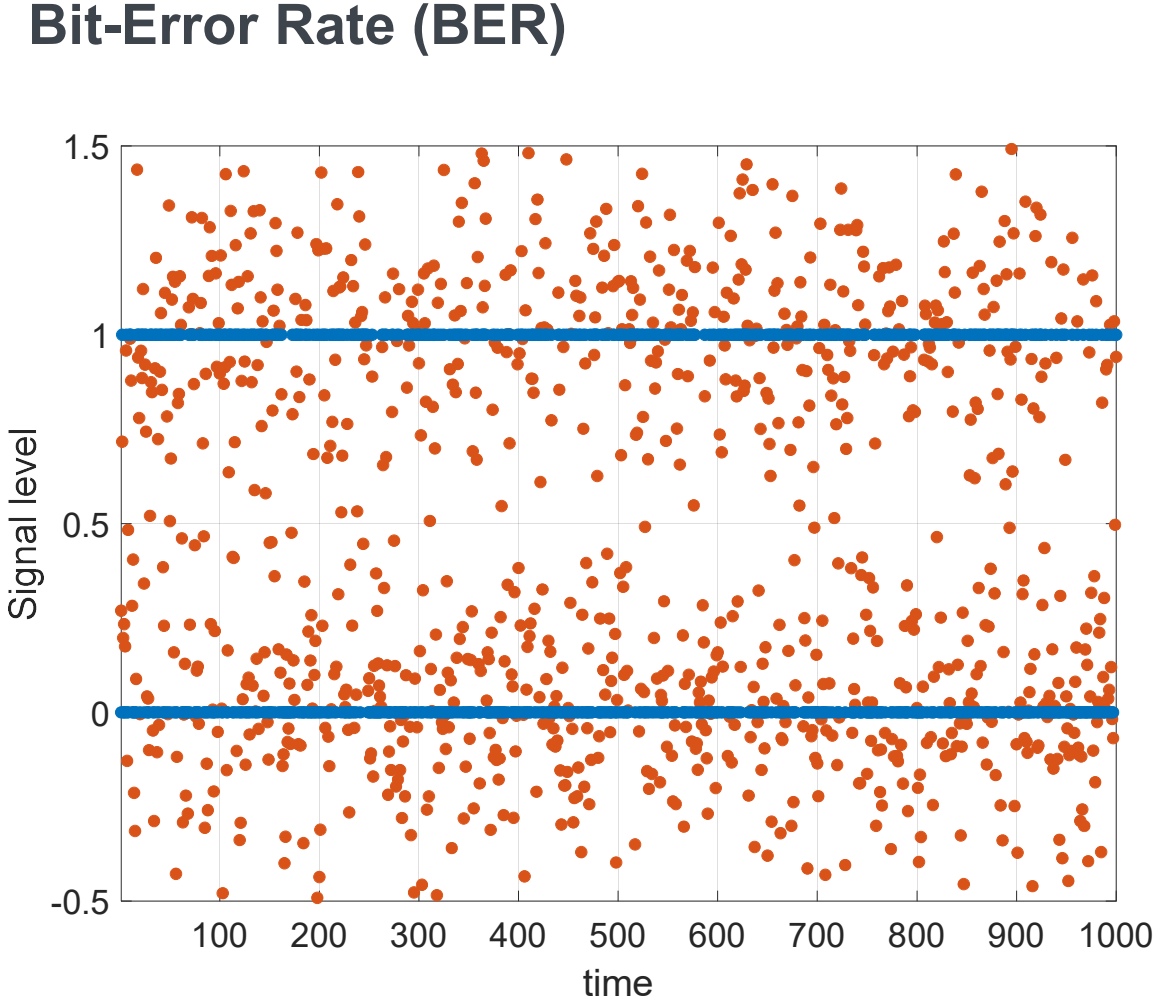


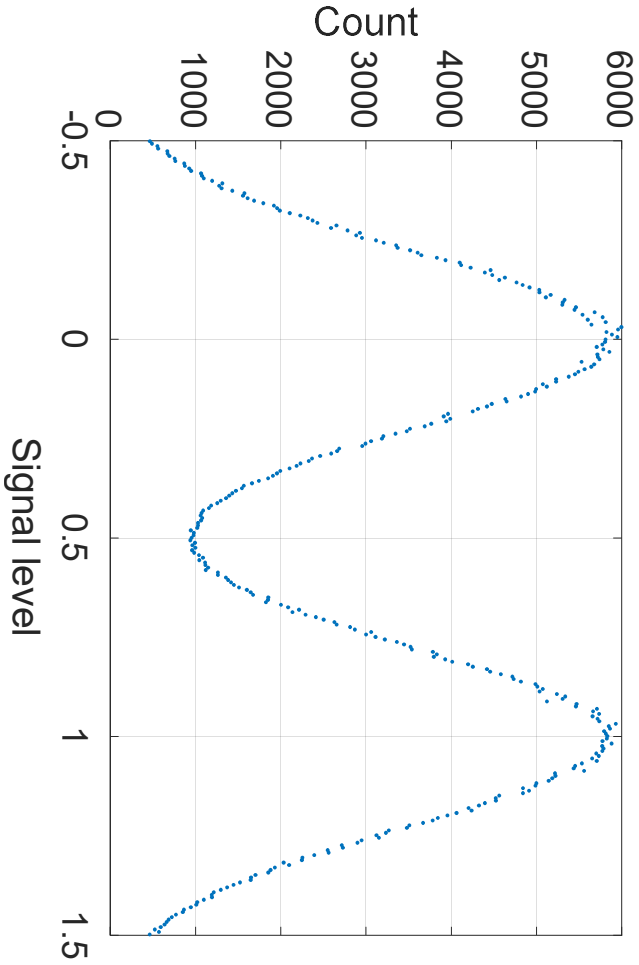
Bit-Error Rate (BER)





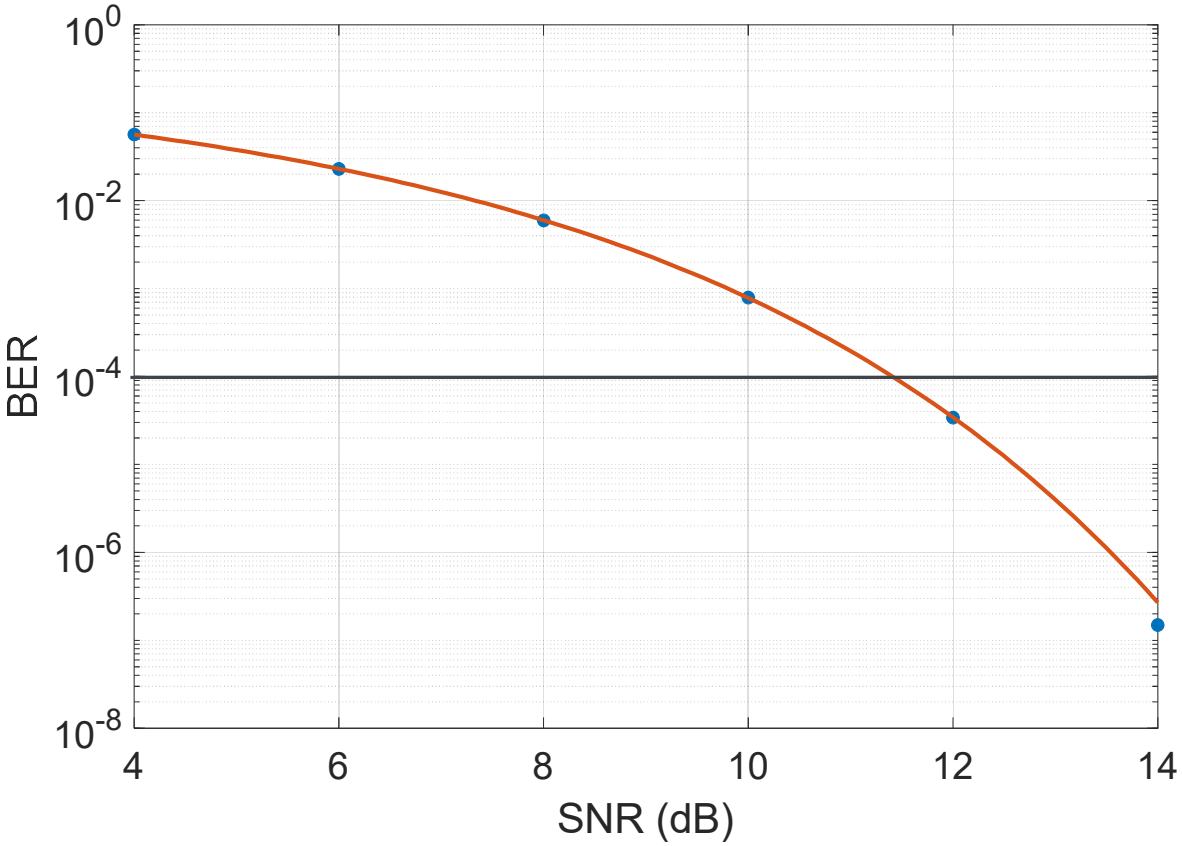
Bit-Error Rate (BER)





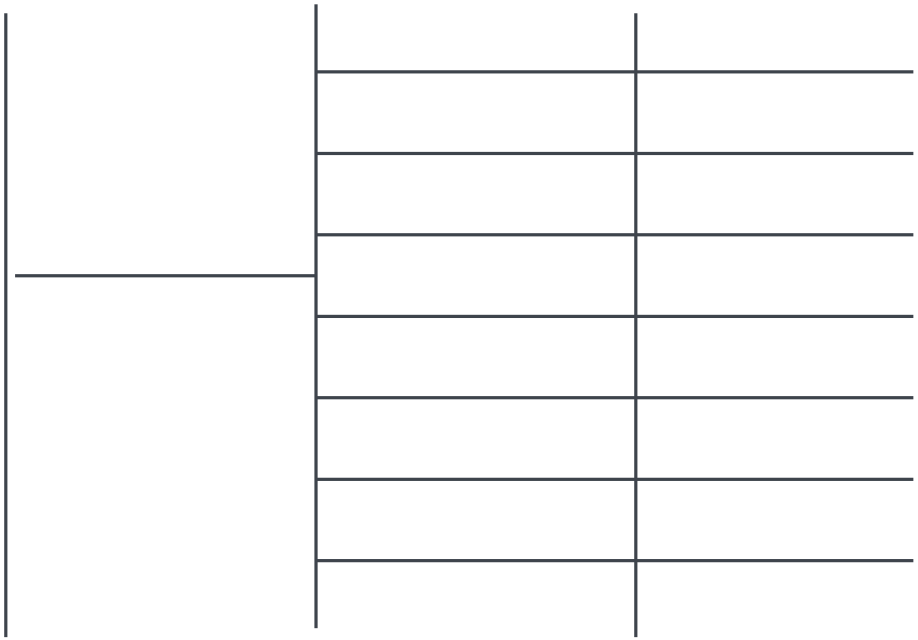
BER vs SNR for OOK

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Forward Error Correction (FEC)

Payload bit Msg



Forward Error Correction (FEC)

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Information-Theoretic Tools for Optical Communications Engineers

(Invited Tutorial)

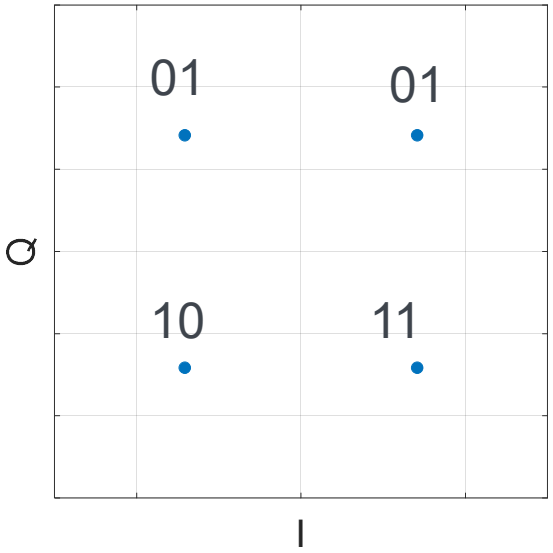
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TABLE I
HARD-DECISION FEC THRESHOLDS OF CERTAIN CODES RECOMMENDED FOR OPTICAL COMMUNICATIONS

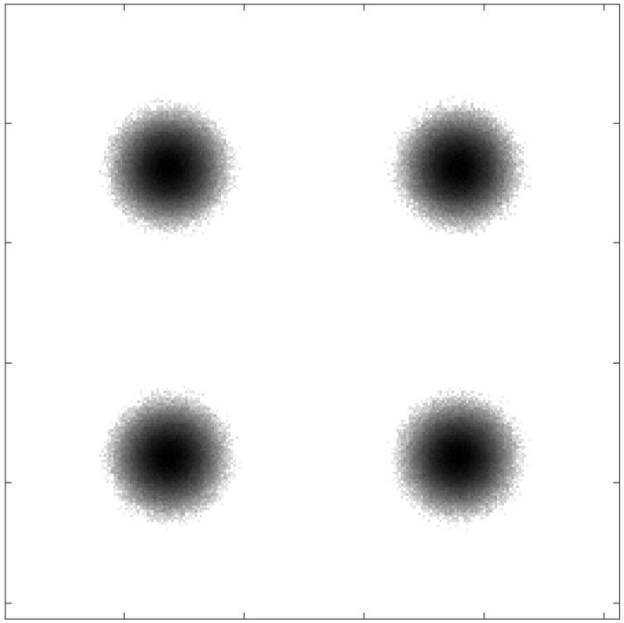
OH	Code type	Source	post-FEC BER	p
2.7%	Reed–Solomon (RS) (528,514) "KR4"	[26][a]	10^{-15}	$2.18 \cdot 10^{-5}$
5.8%	RS(544,514) "KP4"	[26][a]	10^{-15}	$2.26 \cdot 10^{-4}$
6.7%	RS(255,239)	[27][a]	10^{-15}	$8.27 \cdot 10^{-5}$
6.7%	Bose–Chaudhuri–Hocquenghem (BCH)+BCH	[28, App. I.3]	10^{-15}	$3.15 \cdot 10^{-3}$

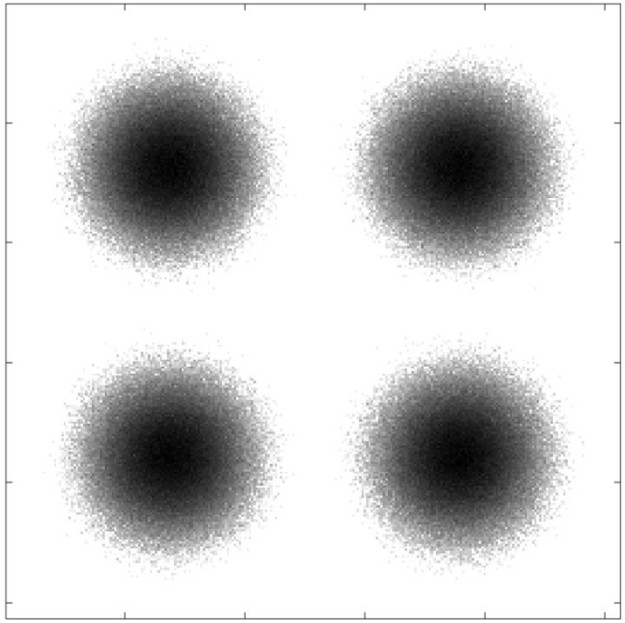
Coherent Signals

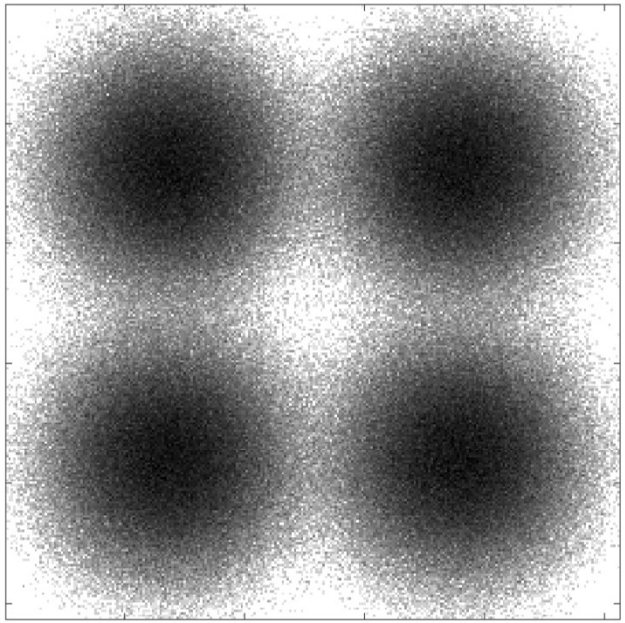


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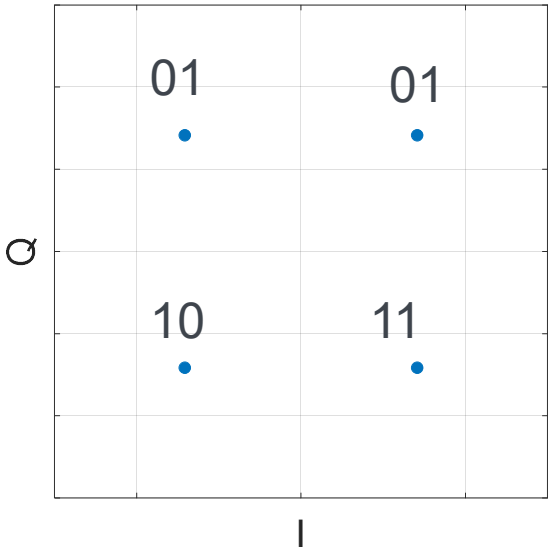


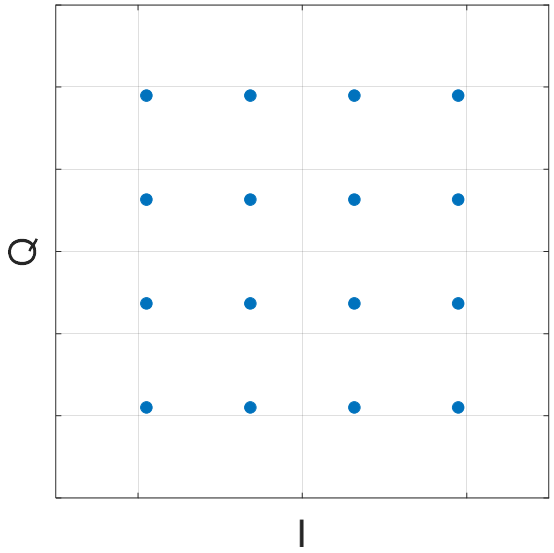


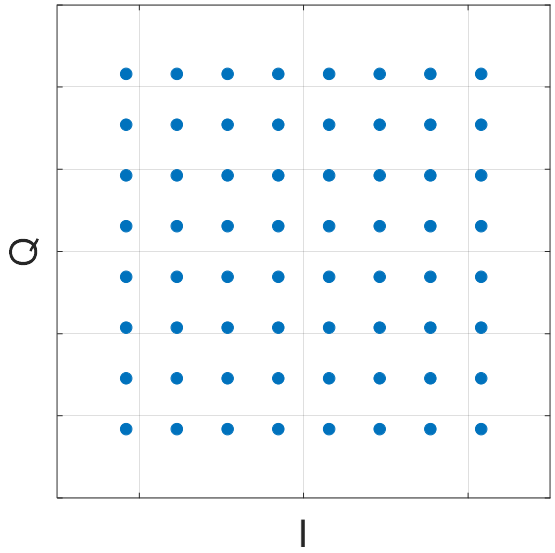


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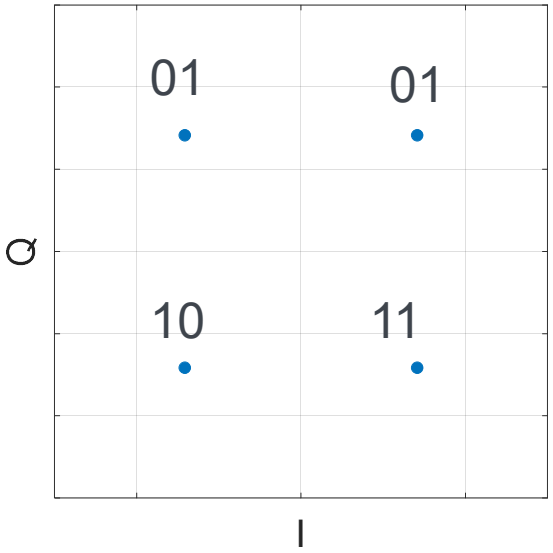


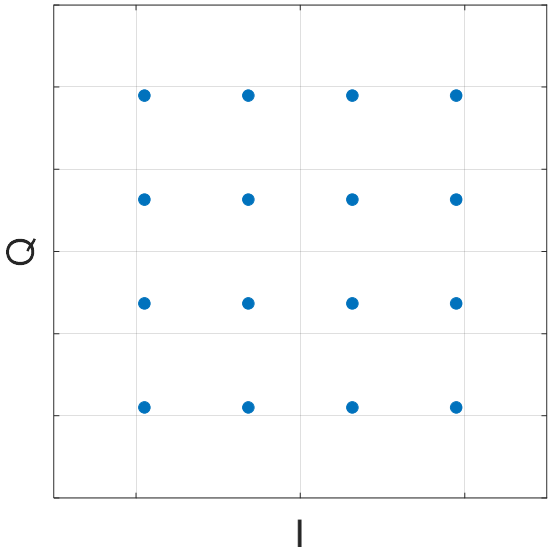


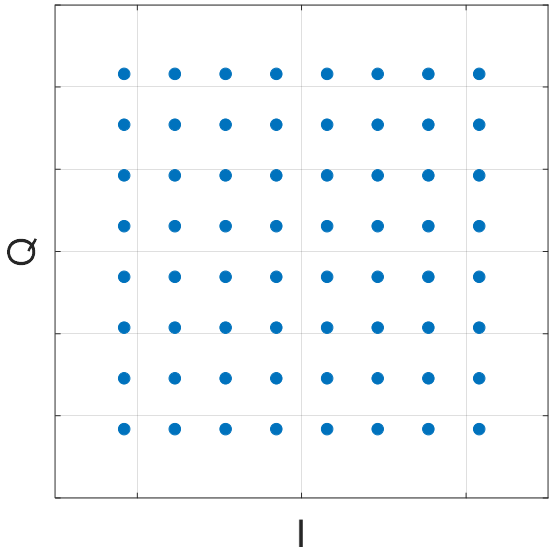


Coherent Signals

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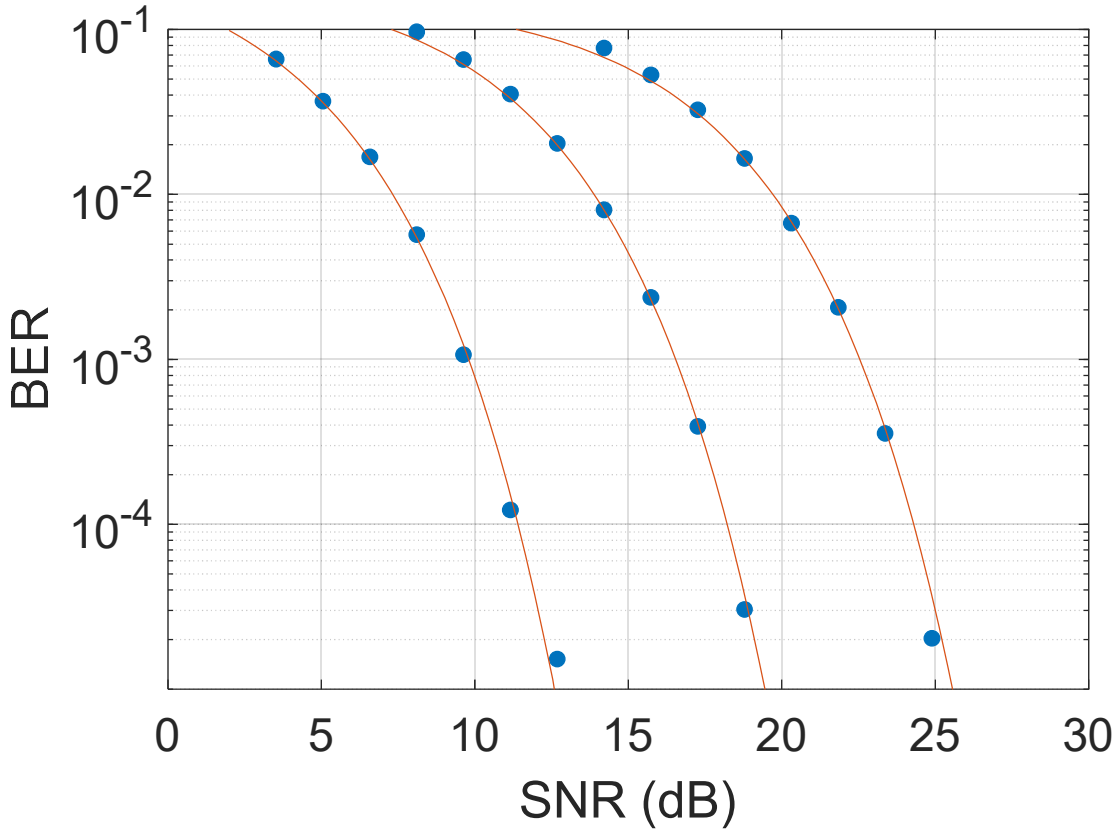






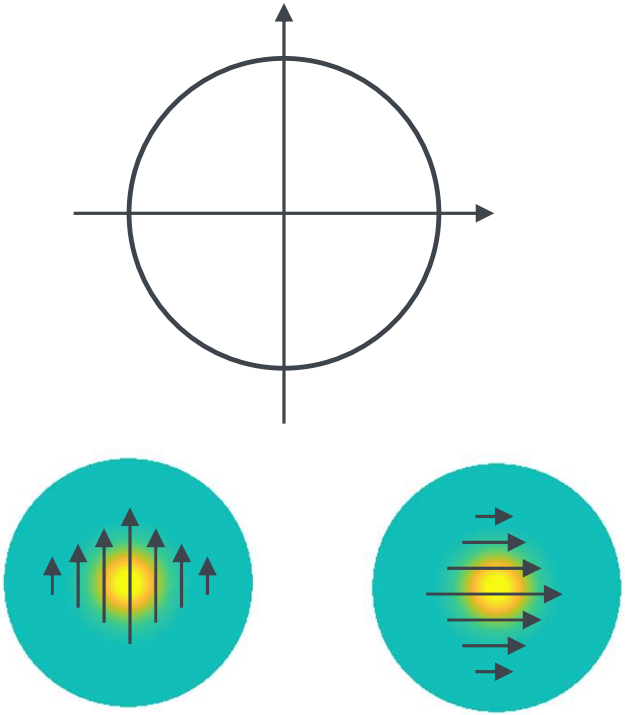
BER vs. SNR

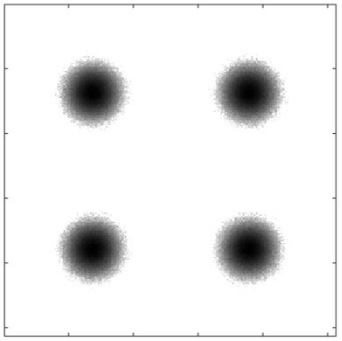
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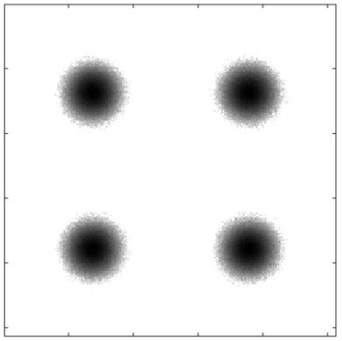


Polarization multiplex (Pol Mux)

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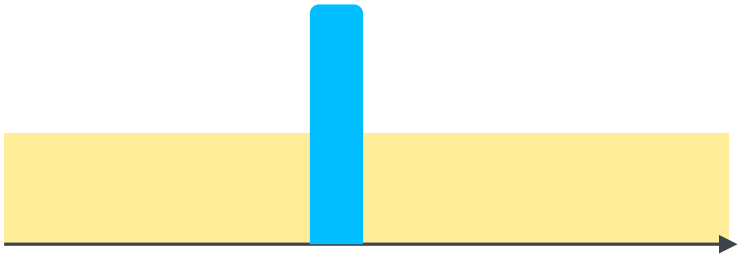






SNR vs. OSNR

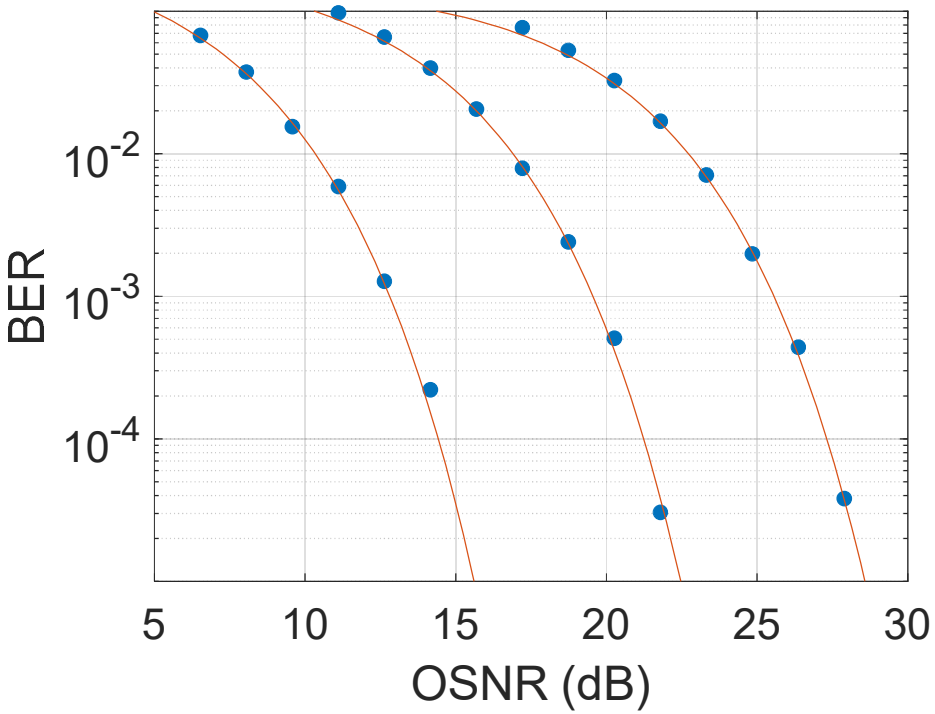
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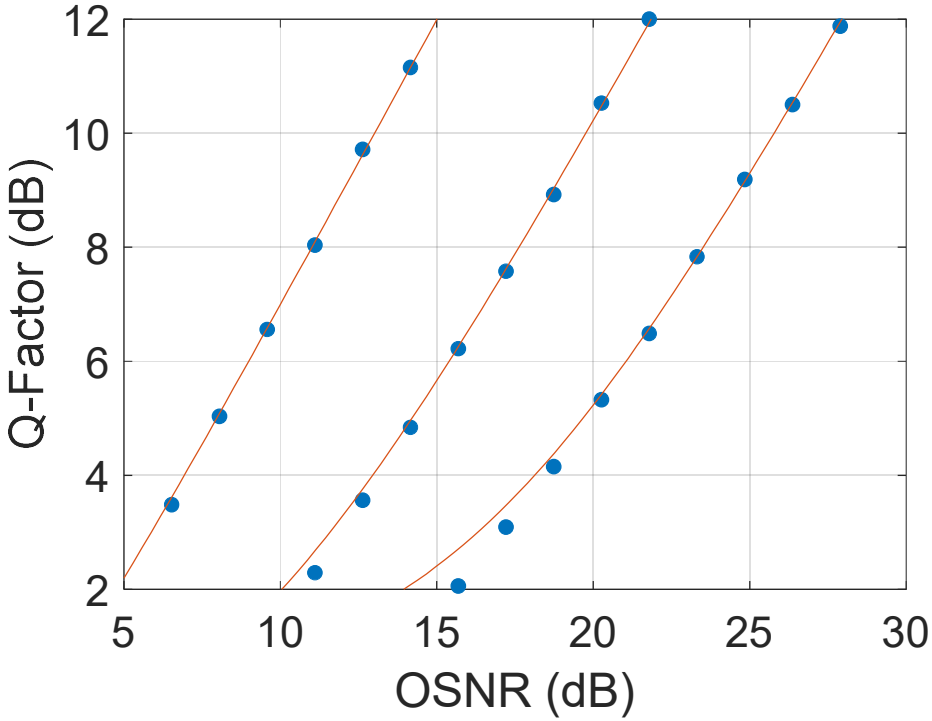




Coherent Signals

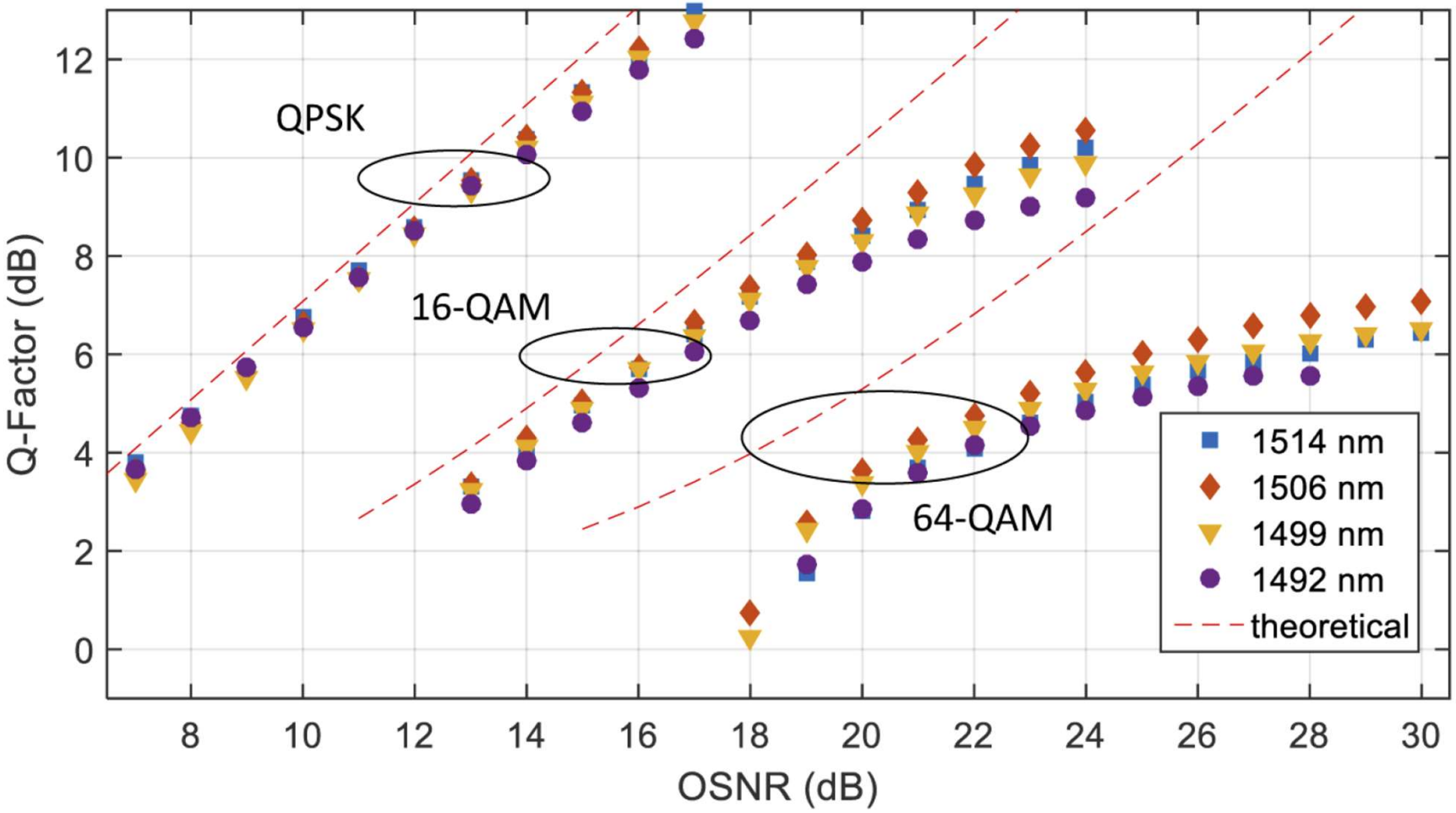
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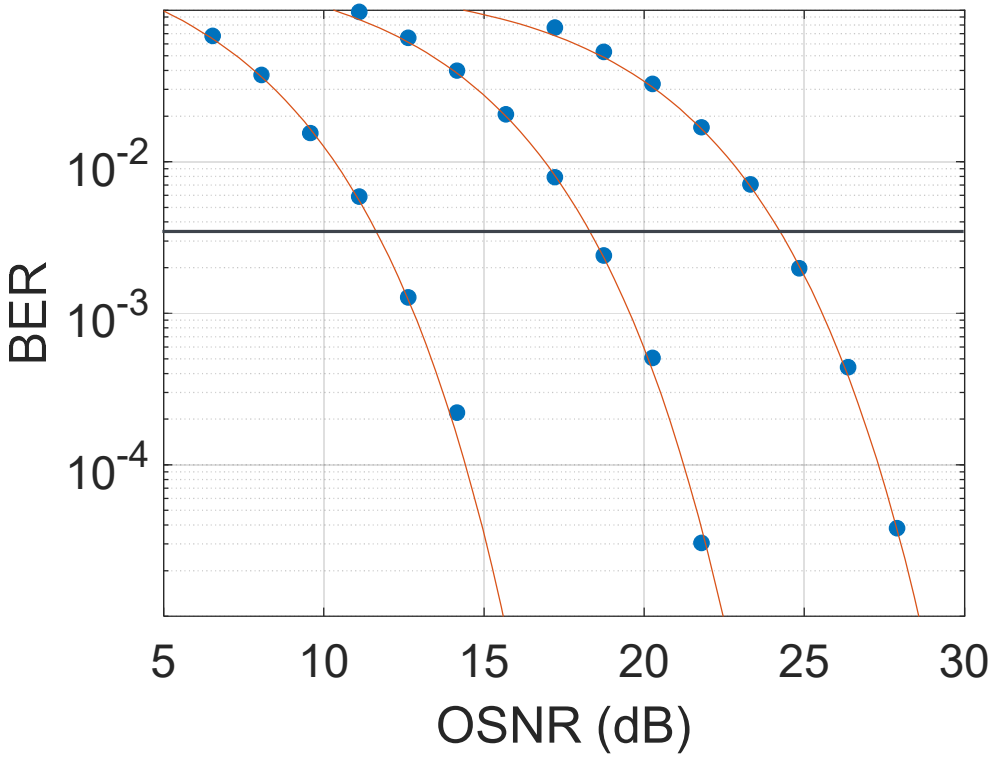
Measured performance

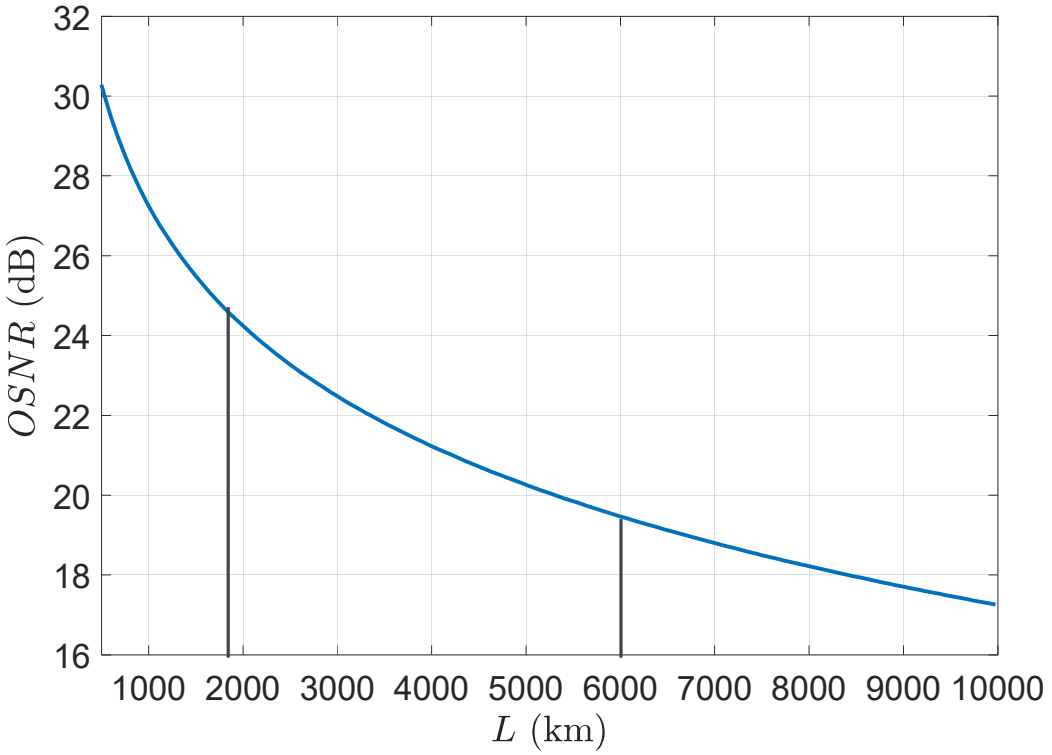
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Long distance transmission

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Pulse propagation in SMF

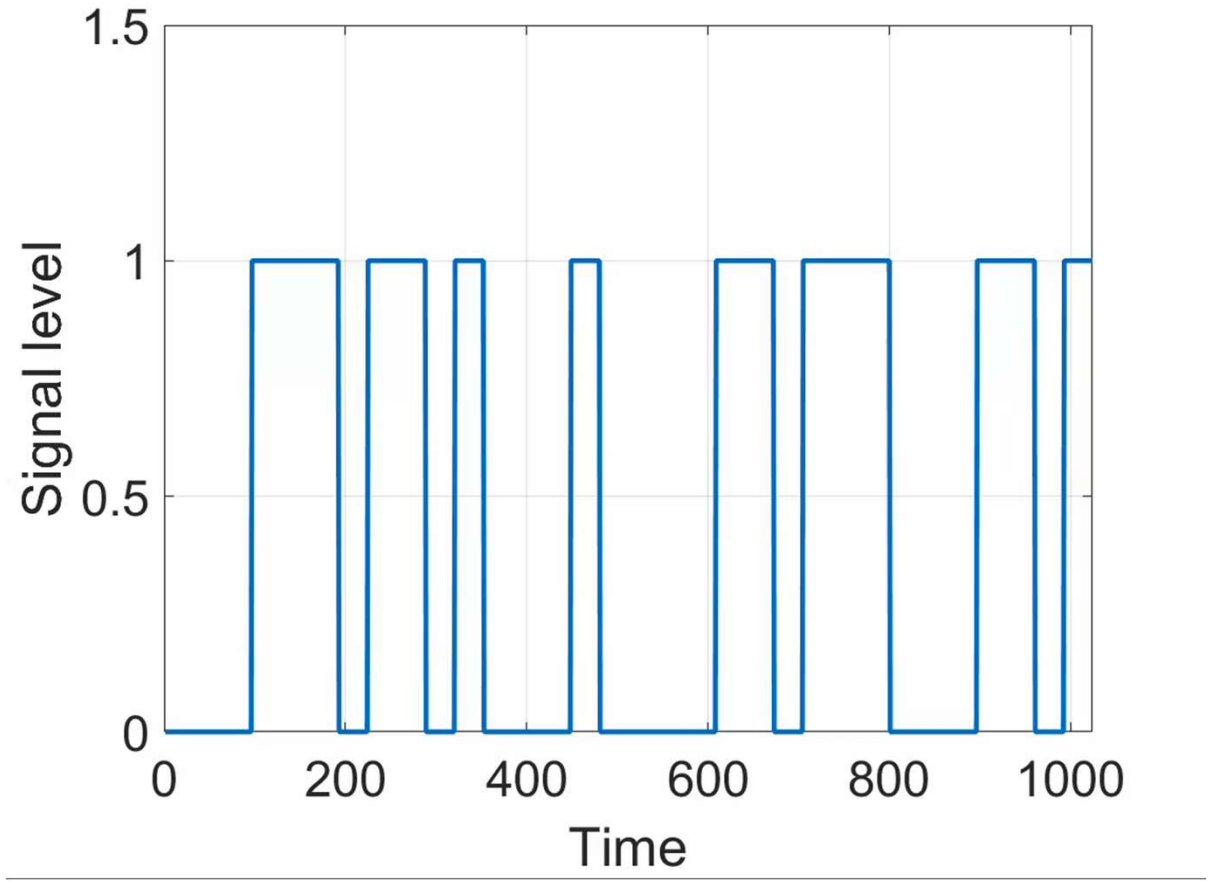
Pulse propagation in SMF

Pulse propagation in SMF

Pulse propagation in SMF

Pulse propagation in SMF

Pulse propagation in SMF


1.5
1
0.5
0
Signal level
0
200
400
600
800
1000
Time

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Thank you!

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